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Magnetostratigraphy and biochronology of the San Timoteo Badlands, southern California, with implications for local Pliocene–Pleistocene tectonic and depositional patterns

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ABSTRACT

The San Timoteo Badlands expose nearly 2000 m of nonmarine sedimentary rocks ideally situated temporally and spatially to address questions regarding major tectonic events that occurred in southern California from latest Miocene to medial Pleistocene time. These events include development of the San Andreas–San Jacinto fault system and uplift of the Transverse Ranges. The badlands sequence also spans three of the four most recent North American Land Mammal "ages" and therefore coincides with important trans-Beringian and Neotropical mammal dispersal events. No other terrestrial sequence in North America records relatively continuous deposition over this interval of time.

Magnetostratigraphy and mammalian biochronology provide a temporal framework that allows greater resolution of the geologic and biotic events recorded within the sequence than previous studies on this region. Data show that the sequence extends from chron C3An.2n, ca. 6.3 Ma, at the lowest part of the section in the study area, to within the Brunhes chron, or younger than 0.78 Ma, near the top. Refined age estimates for geologic and biotic events recorded within the sequence indicate that (1) a Peninsular Ranges basement source south of the badlands region dominated deposition from before 6.3 Ma to about 4.6 Ma, after which time the provenance shifted to a San Gabriel Mountains basement source from the north; (2) deposition of San Gabriel Mountains–type material continued until as recently as ca. 0.70 Ma, although sediments from the San Bernardino Mountains appeared in the northwesternmost area of the badlands ca. 1.5 Ma; (3) the late Hemphillian Mount Eden Local Fauna is dated as 5.6 Ma, compared with previous estimates of 5.0–5.4 Ma; (4) the early Irvingtonian El Casco Local Fauna is 1.3–1.4 Ma; and (5) the Shutt Ranch Local Fauna is ca. 0.7–0.99 Ma. Paleomagnetic data show that the badlands region may be slightly rotated in a counterclockwise direction, contrary to previous models predicting clockwise rotation in right-lateral shear zones.

INTRODUCTION

The Mount Eden and San Timoteo formations of the San Timoteo Badlands (hereafter referred to as the Badlands) represent the only known nonmarine sequence in North America that records relatively continuous deposition from latest Miocene to medial Pleistocene time. Because of this and the location of the Badlands in the tectonically complex and active region of southern California between the San Bernardino basin and San Gorgonio Pass (Fig. 1A), the area has figured prominently in studies addressing the paleogeographic evolution of the southern San Andreas fault zone, particularly regarding inception of the San Jacinto fault zone and uplift of the San Bernardino Mountains (Matti and Morton, 1993; Morton and Matti, 1993a, 1993b). Deposition of this sequence also spanned three of the four most recent North American Land Mammal "ages," the Hemphillian, Blancan, and Irvingtonian, and therefore coincided with important trans-Beringian dispersal events driven primarily by alternating glacial-interglacial climatic cycles, as well as the Neotropical wave of immigration known as the Great American Interchange that resulted from development of the Panamanian isthmus (e.g., Webb, 1985; Repenning, 1987).

Despite the significance of the Badlands as a focal point for tectonic studies in this region, the age of the sequence has not been calibrated by modern geochronological techniques. Previous workers (e.g. Matti and Morton, 1975; Morton and Matti, 1993a) have relied on relative age-based data for temporal control rather than on numeric or correlated-age stratigraphy (sensu Rosholt et al., 1991, p. 46). One of the major aims of this paper, therefore, is to present a refined chronostratigraphic analysis that provides a temporal framework to aid future studies regarding local tectonic, paleobiologic, and paleoclimatic events.

This paper details the development of that framework by using paleomagnetic and mammalian biochronological data. It modifies and expands upon the magnetostratigraphic foundation developed in Hehn et al. (1996) for the lower part of the Badlands section and presents new magneto-biostratigraphic data for the upper part. The new framework is then used to reexamine, and refine where necessary, the timing of local tectonic and sedimentological events recorded within the sequence. There are 22 major reversal boundaries in the global magnetic polarity time scale (MPTS) over the ~6 m.y. represented by the Badlands sequence, thus allowing a potential geologic resolution at a scale of about 0.27 m.y.

This paper also reexamines the counterclockwise tectonic rotation of the Badlands region reported in Hehn et al. (1996). It briefly attempts to resolve the discrepancy between the paleomagnetic data and previous models based on seismic studies that predict clockwise vertical axis rotation in this region. Also presented are new age estimates for fossil mammal-bearing localities collected in the Badlands during the early part of the twentieth century, such as the Mount Eden and El Casco local faunas (Frick, 1921, 1937).

HISTORICAL BACKGROUND

From 1916 to 1917 Childs Frick conducted a paleontological reconnaissance of the Badlands area as part of a University of California study

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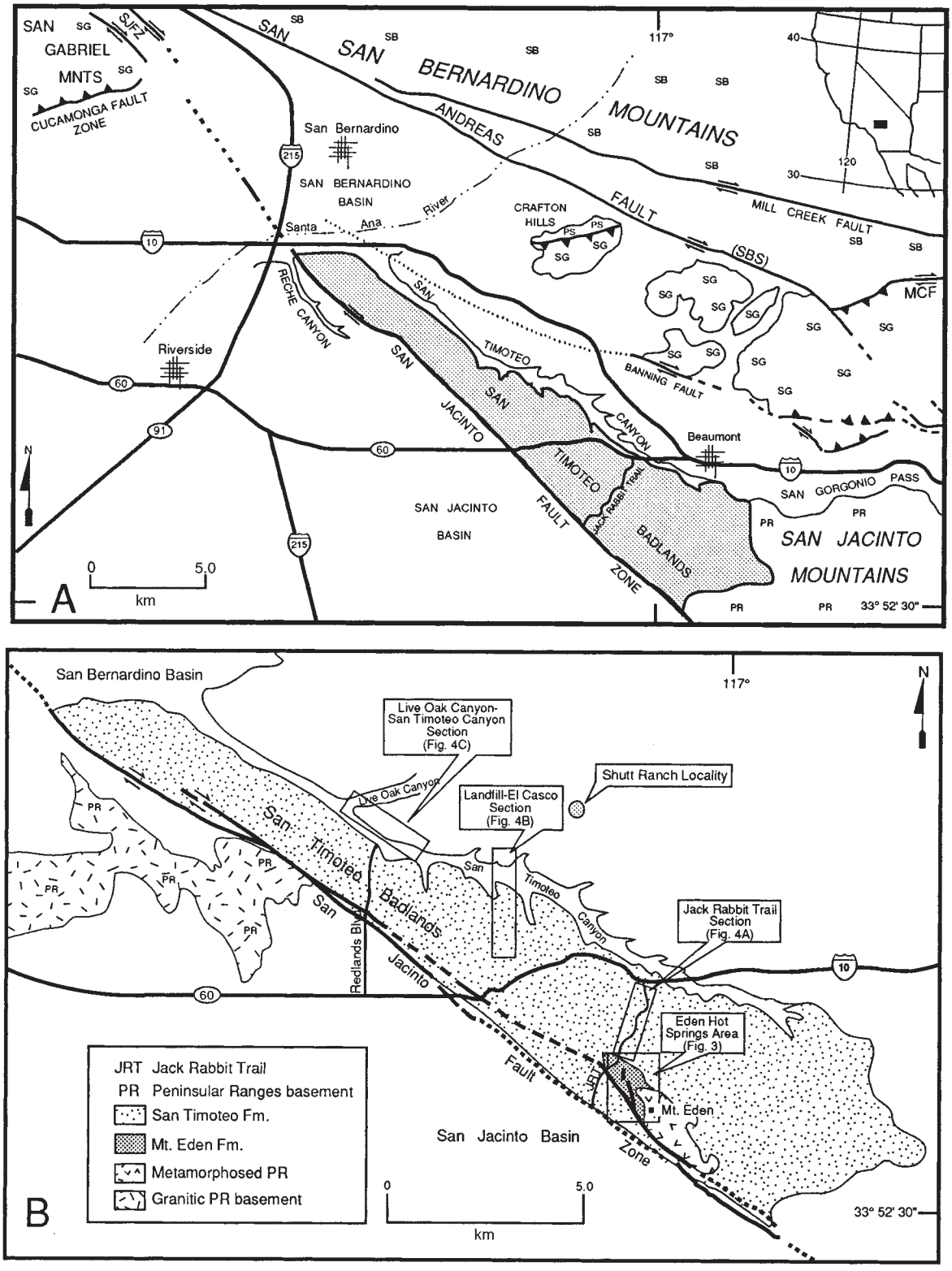


Figure 1. (A) Location and regional setting of the San Timoteo Badlands. MCF—Mission Creek fault; PR—Peninsular Ranges–type basement; PS—Pelona Schist; SB—San Bernardino Mountains–type basement; SBS—San Bernardino strand; SG—San Gabriel Mountains–type basement; SJFZ—San Jacinto fault zone. (B) Generalized geology of the Badlands region with insets showing locations and text figure references of composite stratigraphic sections. Base is from U.S. Geological Survey San Bernardino sheet.

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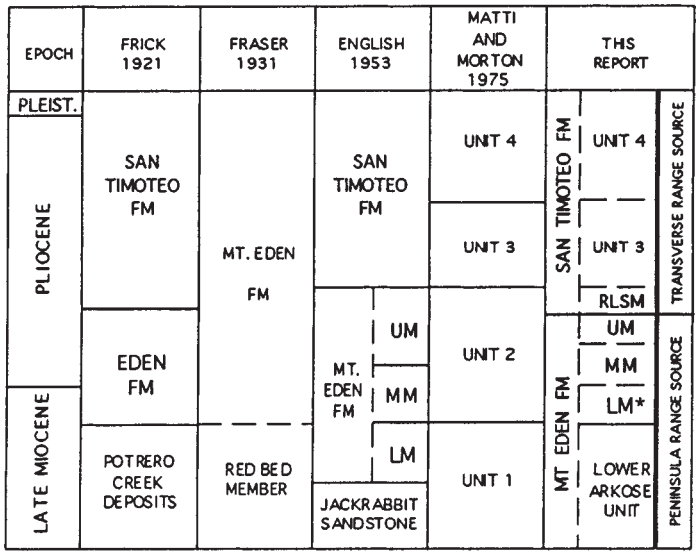


Figure 2. Historical and present nomenclature for stratigraphic units of the San Timoteo Badlands. LM, MM, UM—lower, middle, and upper members of the Mount Eden formation; RLSM—ripple-laminated sandstone member of the San Timoteo formation; asterisk denotes stratigraphic position of the Mount Eden Local Fauna.

on the "geologic and faunal history of the Pacific coast ..." (Frick, 1921, p. 279). Based on identification of the mostly large mammal fossils, Frick (1921, p. 283) concluded that the lower part of the sequence, his "Eden beds" or "Eden formation" (Fig. 2), represented the "upper division of the Lower Pliocene." In a later study, Frick (1933, p. 516) considered the age as "Uppermost Pliocene." Axelrod (1934, 1950) reached a similar conclusion on the basis of plant fossils collected in the Badlands.

May and Repenning (1982) were the first to use magnetostratigraphy in the Badlands, noting the importance of the sequence as a temporal record of San Andreas slip histories and Transverse Ranges uplift. They concluded that the Mount Eden local fauna was between 5.4 and 5.0 Ma, based on horse, proboscidean, and rodent fossils, coupled with the paleomagnetic signature of the strata from which the rodent fossils were thought to be recovered.

Reynolds and Reeder (1991) provided a general overview of the lithology, structure, and paleontology of the Badlands, and added further constraints on the age. Based on small mammal fossils (identified, in part, by C. A. Repenning) from three localities found high in the San Timoteo formation (discussed in the following), Reynolds and Reeder (1991) concluded that deposition continued until about 1.3 Ma. This study indicates that deposition may have continued until about 0.7 Ma.

GEOGRAPHIC AND GEOLOGIC SETTING

The Badlands sequence was deposited into a basin located between what are now the San Andreas and San Jacinto fault zones (Fig. 1). Movement along the San Jacinto fault zone over the past 1.5 m.y. uplifted the area, resulting in a predominantly anticlinal structure that gently plunges to the northwest (Morton and Matti, 1993b). The geologically oldest part of the sequence crops out in the southeastern part of the Badlands where strata nonconformably overlie the northwestern part of the San Jacinto Mountains block. From there, the Badlands extend northwest for ~30 km to the southern extent of the San Bernardino basin (Fig. 1, A and B). Most of the Badlands region is between the San Jacinto fault zone, to the southwest, and San Timoteo Canyon, to the northeast, although the uppermost part of the section is exposed north of the canyon. About 10 km north of the Badlands is the San Bernardino strand of the San Andreas fault zone and the southern front of the San Bernardino Mountains. Between the San Andreas fault and San Timoteo Canyon is the Banning fault, along which Matti et al. (1992) reported no evidence of Quaternary offset.

This study focuses on an area from low in the sequence near Mount Eden in S24, T3S, R2W, of the El Casco U.S. Geological Survey (USGS) 7.5′ quadrangle, Riverside County, northwest to near the top of the sequence in Live Oak Canyon on the north side of San Timoteo Canyon in S11, T2S, R3W, of the Redlands USGS 7.5′ quadrangle, San Bernardino County (Fig. 1B). Work was confined to the northern limb of the main anticline in the Badlands to avoid structural complications on the south limb caused by deformation along the San Jacinto fault zone.

Lithostratigraphic Framework

The stratigraphy of the Badlands sequence is presented as three composite sections constructed from numerous shorter sampling transects (Figs. 1B, 3, and 4, A–C). From base to top they are: (1) the Eden Hot Springs–Jack Rabbit Trail composite section, which includes arkosic beds at the base of the sequence that represent the upper part of Frick's (1921) Potrero Creek deposits, overlying fluvio-lacustrine deposits that typify the Mount Eden formation, and the lower part of the mostly fluvial San Timoteo formation; (2) the Riverside County Landfill–El Casco composite section, which is entirely within the San Timoteo formation; and (3) the San Timoteo Canyon–Live Oak Canyon composite section within the uppermost San Timoteo formation. The Mount Eden and San Timoteo "formations" will be formally named and defined with publication of the geology of the USGS El Casco 7.5′ quadrangle (J. Matti, 1996, personal commun.).

Eden Hot Springs–Jack Rabbit Trail Composite Section. The stratigraphically lowest part of the section consists primarily of a fluvially derived, coarse-grained, poorly sorted, variegated arkose. Clast types indicate derivation from Peninsular Range basement sources generally to the south of the Badlands (Matti and Morton, 1975, 1993). These strata correlate to the upper part of Frick's (1921) Potrero Creek deposits and Fraser's (1931) Red Bed Member of the Mount Eden Formation. English (1953) referred to these beds as the Jack Rabbit Sandstone and the "lower member of the Mount Eden Formation," and Matti and Morton (1975) labeled them "unit 1." Here, these deposits are referred to as "the lower arkose unit" and are considered to be a member of the Mount Eden formation (Fig. 2). Absence of fossils from this unit has led to speculation regarding its age, although it is generally considered to be late Miocene age (Frick, 1921; Fraser, 1931). Confirmation of a late Miocene age is presented elsewhere in this report.

The lower arkose unit is exposed at the base of the Eden Hot Springs measured section and in the hinge of the anticline along Jack Rabbit Trail about 0.5 km west of the Eden Hot Springs area (Fig. 3). Jack Rabbit Trail is an unmaintained road that extends northward across the anticline nearly perpendicular to strike, providing roadcuts that allowed measurement of the section and sampling for fossils and paleomagnetic analysis (Figs. 1, 3, and 4A). Evaluation of faults along Jack Rabbit Trail determined that parts of the section may have been cut out, but not repeated. Paleomagnetic sites 131–133 were collected in the lower arkose unit.

Conformably overlying and interfingering with the lower arkose unit are the gray, fluvio-lacustrine mudstones and sandstones more typical of the Mount Eden formation (see Fraser, 1931, p. 512–513). English (1953) divided the Mount Eden formation into lower, middle, and upper members. His lower member may be equivalent to the upper part of Frick's Potrero Creek deposits (Fig. 2) and his middle and upper members are equivalent to

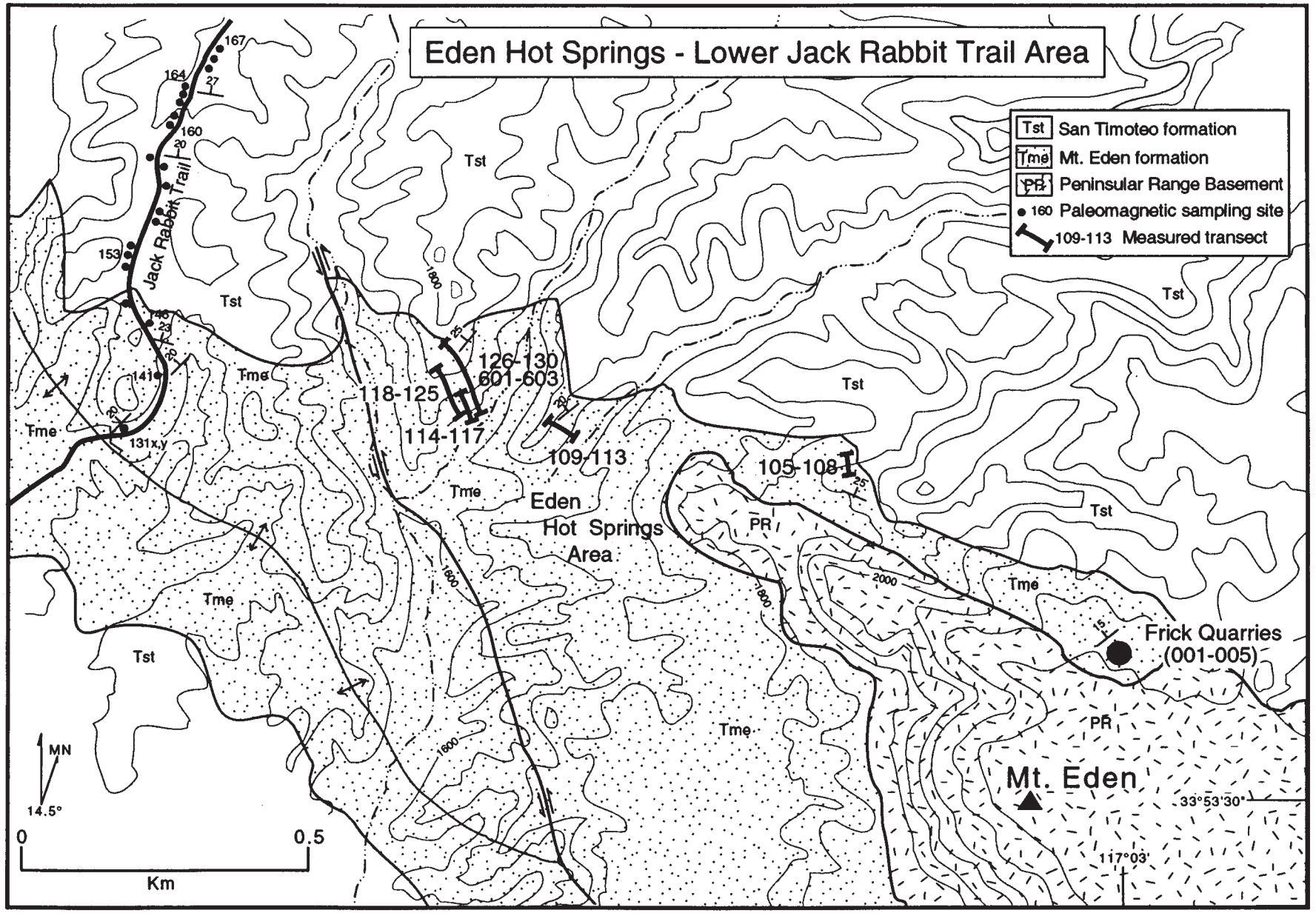


Figure 3. Eden Hot Springs–lower Jack Rabbit Trail area showing general geology plus transects and sites sampled for paleomagnetic analysis. El Casco U.S. Geological Survey 7.5′ topographic base. Map has 50 ft contour interval; 50 ft = 15.24 m.

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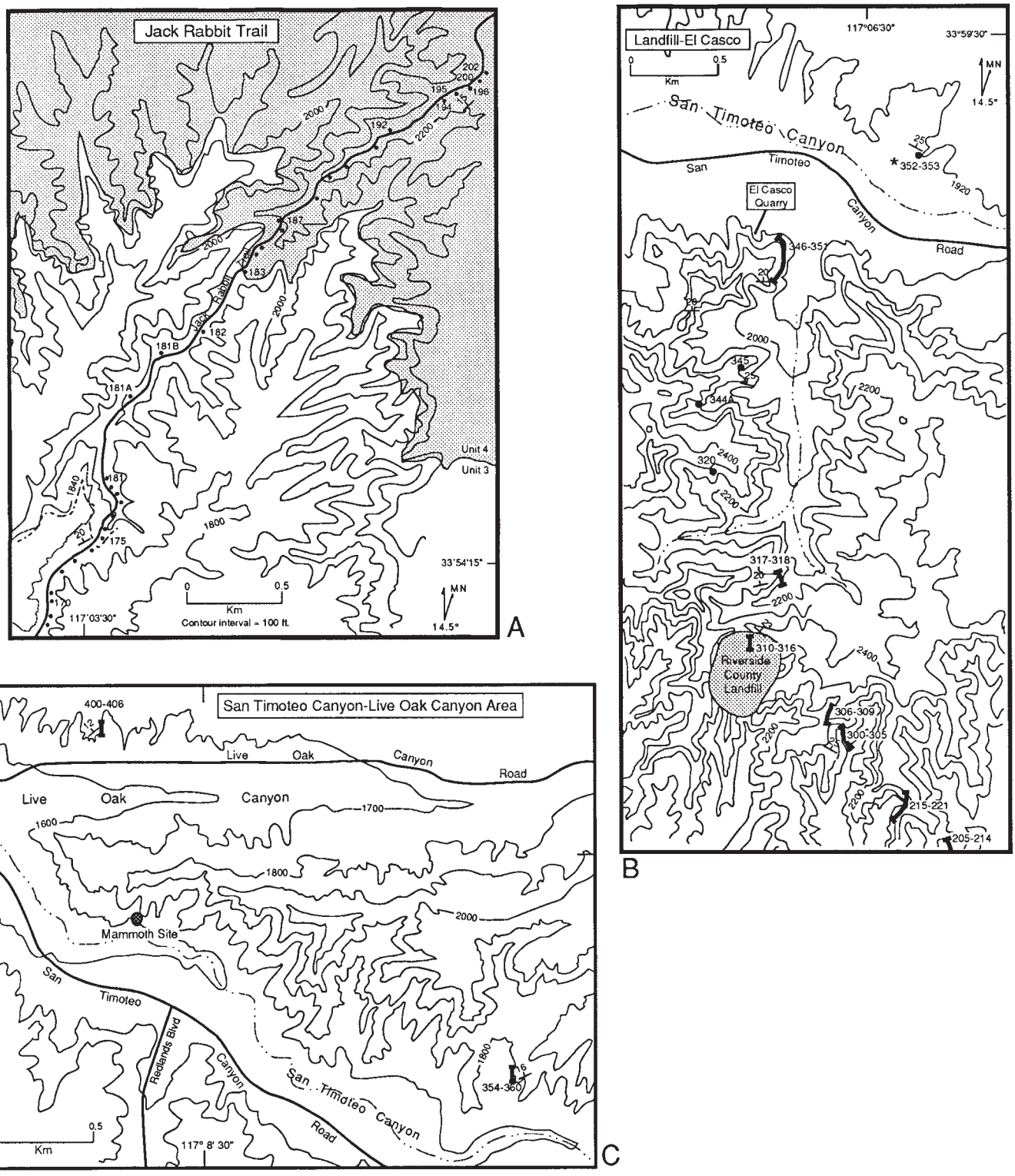


Figure 4. (A) Location of sites sampled for paleomagnetic analysis along remainder of Jack Rabbit Trail section. Stippled area denotes unit 4 (Matti and Morton, 1975; see text) of the San Timoteo formation. Dashed line between sites 174 and 175 marks a persistent limestone bed. El Casco U.S. Geological Survey (USGS) 7.5′ topographic base. (B) Locations of transects and sites sampled for paleomagnetic analysis that comprise the Riverside County Landfill–El Casco composite section. The two transects that include sites 300–309 are combined as transect 3 in the text and in Figures 10 and 13. Sites 320, 344, and 345 comprise transect 6. Asterisk denotes transect 1 of the San Timoteo Canyon–Live Oak Canyon composite section. Transect 2 of the same is located about 1.6 km northwest of transect 1 (see Fig. 4C). El Casco USGS 7.5′ topographic base. (C) Locations of transects 2 and 3 of the San Timoteo Canyon–Live Oak Canyon composite section and Mammoth site (see discussion in text). Sunnymead and Redlands USGS 7.5′ topographic base. All maps have 100 ft contour intervals; 100 ft = 30.48 m.

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unit 2 of Matti and Morton (1975). Here, the lower arkose unit is considered the basal member of the Mount Eden formation and the overlying fluviolacustrine strata are divided into three additional members also termed "lower," "middle," and "upper" (Fig. 2). The lower member consists of grayish calcareous sandstones, siltstones, and mudstones with prominent limestone beds. It was from this member, on the northwest side of Mount Eden, that the bulk of the Mount Eden local fauna was obtained (the "Eden ledges" locality of Frick, 1921, p. 339). The middle member consists primarily of grayish and brownish muddy sandstone with occasional limestone beds. The upper member is characterized by dark grayish-green mudstone interspersed with occasional indurated muddy sandstone, siltstone, and claystone beds; limestone beds are absent. Hills underlain by this member weather into a subdued topography. Paleomagnetic sites 1–5, 100–130, 134–143B, and 601–603 were collected in the Mount Eden formation (Fig. 3).

Overlying and interfingering with the lacustrine mudstones of the upper Mount Eden formation is a tan, fine-grained sandstone here termed the "ripple-laminated sandstone" from which paleomagnetic sites 144–150 were collected. Rare clasts of San Gabriel Mountains–type basement define this unit as lowermost San Timoteo formation (J. Matti, 1996, personal commun.). Lenticular bedding, ripple marks, soft-sediment deformation structures, and occasional beds of dark shale to 1 m thick characterize this member. The distinctive nature of this sandstone allows confident correlation between the Eden Hot Springs area and Jack Rabbit Trail (Fig. 3). Overlying this unit are the coarser-grained, fluvially derived deposits more typical of the San Timoteo formation. From these strata along Jack Rabbit Trail paleomagnetic sites 151–196 and 200–202 were collected (Figs. 3, and 4A).

The San Timoteo formation is by far the most extensive unit in the Badlands. It may be as thick as 2000 m, although accurate measurement is difficult due to vegetation cover, structural complexity, rarity of laterally continuous marker beds, and discontinuous exposure. Clast types indicate derivation from the eastern San Gabriel Mountains and the present location of the San Bernardino basin north and northwest of the Badlands (Matti and Morton, 1993; Morton and Matti, 1993a). The first prominent bed of such clasts is seen along Jack Rabbit Trail only a few meters above the ripple-laminated sandstone near paleomagnetic site 153 (Fig. 3). Above the ripple-laminated sandstone, the San Timoteo formation consists primarily of poorly indurated sandstone and sandy mudstone with occasional conglomeratic beds that in many places scour underlying strata. Matti and Morton (1975) labeled this part of the section unit 3 (Fig. 2). Although there is no distinct boundary, the lithologic character changes upsection in the vicinity of sites 183–186 (Fig. 4A) to a generally coarser-grained sequence characterized by thick, sheet-like conglomeratic beds. Matti and Morton (1975) labeled this part of the sequence unit 4. Lack of crossbedding and pervasive lateral facies changes support their conclusions that deposition occurred primarily as a result of intermittent braided stream sheetwash. Many of the conglomeratic beds grade upward into finer fluvial layers which, in turn, grade into reddish paleosols. Stacked sequences of these fining-upward packages characterize the overall stratigraphy of the San Timoteo formation.

Riverside County Landfill–El Casco Composite Section. The Landfill–El Casco composite section is about 6.5 km northwest of the Jack Rabbit Trail section and begins about 0.5 km north of California Highway 60 (Figs. 1B and 4B). Although direct lithostratigraphic correlation between the two sections is not possible due to facies changes and lack of distinct marker beds, the section appears to be confined to unit 4. Paleomagnetic and biostratigraphic analyses indicate that the two sections partly overlap (discussed in the following). Near the top of this section, about 50 m stratigraphically below the El Casco fossil locality, the lithology changes from the predominantly reddish to orange sandstone and thick conglomeratic beds to overlying grayish, finer-grained, less conglomeratic strata.

San Timoteo Canyon–Live Oak Canyon Composite Section. This section begins across San Timoteo Canyon from the top of the Landfill–El Casco section and is about 300 m stratigraphically higher, based on projections of bedding attitudes. The first of three transects is in a small exposure of siltsone about 5 m thick, from which sites 352–353 were sampled (Fig. 4B). Transect 2 (Fig. 4C) is about 1.6 km northwest of transect 1 and includes about 24 m of predominantly fine-grained sandstone to siltstone with a conglomeratic bed at the base and a calcareous mudstone near the top. Sites 354–360 were collected in transect 2. Sites 400–406 were collected in the stratigraphically highest measured section in the study area, about 1.8 km northwest of transect 2 on the north side of the western end of Live Oak Canyon (Fig. 4C). This section is ~100 m thick and consists of siltstone to fine-grained sandstone in the lower part. The upper part is mostly covered and was not sampled. At the top of this section is a preserved geomorphic surface designated Qvoa[1] by Matti et al. (1992).

METHODS

Correlations and age determinations presented within are based on the MPTS of Berggren et al. (1995); correlations in previous literature are accordingly updated. For the interval of time over which this study is concerned, the past 6 m.y., Berggren et al. (1995, p. 138) concluded that magnetochron boundary age estimates reported by Mankinen and Dalrymple (1979) were about 5%–7% too young. Also in this report "Ma" refers to the age of an event in millions of years referenced to a radioisotopically calibrated time scale; "m.y." refers to one millions years in duration or years ago, as does "k.y." for one thousand years; and the terms "Fauna" and "Local Fauna" follow usage in Tedford et al. (1987).

Fossil Collecting

Despite the presence of the Mount Eden, El Casco, and Shutt Ranch local faunas, fossils are scarce throughout the Badlands. Accordingly, nearly 23 000 kg of matrix were collected from 30 localities for microvertebrate processing. Most sites were chosen on the basis of lithology, with a focus on paleosols, although two sites were chosen because large mammal remains were found. Matrix collected from the same level as a paleomagnetic sample was given the number of that paleomagnetic site (e.g., site 164); matrix collected from a level some measured distance above a paleomagnetic locality was given the number of the nearest paleomagnetic site stratigraphically below it, plus a letter to denote whether it was the first locality sampled above the paleomagnetic site (e.g., 164A), the second (164B), and so on. Matrix was loaded into 20 mesh screens then agitated to wash away the fine particle fraction, leaving a concentrate that was dried then sorted under a 1.5x circular lighted magnifier. Detailed systematic treatment of all taxa collected in the course of this study was presented in Albright (1997).

Paleomagnetic Sampling and Measurement

At least three oriented samples were collected from each site following the long-accepted sampling procedure noted in, e.g., Lindsay et al. (1987). From 20 measured sections, 198 sections were sampled, all on the north limb of the anticline, from the lower arkose unit near the hinge of the anticline along Jack Rabbit Trail, up to the ca. 500 k.y. soil in Live Oak Canyon near the highest part of the San Timoteo formation. The stratigraphic sampling interval averaged ~2–4 m between sites, but varied widely in order to obtain suitable lithologies. In figures and text, site numbers followed by the letters A or B (e.g., 159A) refer to localities sampled some measured distance stratigraphically above the numbered locality but below the next consecutively numbered locality. Those numbers followed by the letter R (e.g.,

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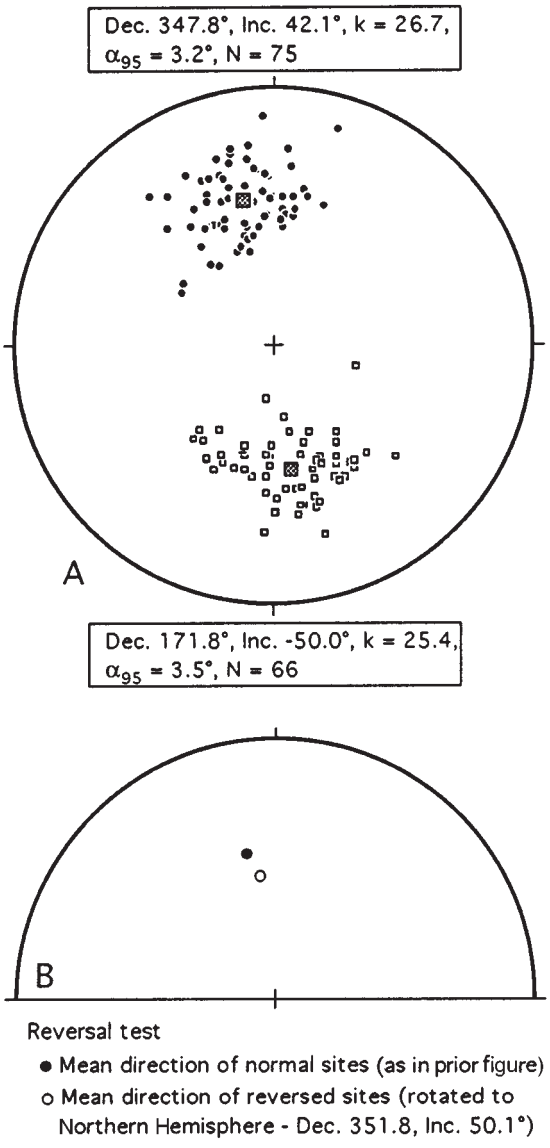


Figure 5. (A) Equal-area stereo plot with mean values and Fisher statistical parameters for all class I sites in the Mount Eden and San Timoteo formations. Solid circles—normal polarity sites; open squares—reversed polarity sites; larger shaded squares—mean directions. Not corrected for declination anomaly due to plunging anticline. (B) Reversals test showing mean direction of all class I sites, reversed sites inverted to Northern Hemisphere.

160R) refer to sites that were resampled at, or very near, the original sampling locality due to loss of the original sample or ambiguous first-run results. Bedding orientations were measured at or near most sites for structural corrections, and dips were typically less than 25° (Figs. 3 and 4).

Most samples collected in the Eden Hot Springs area, along Jack Rabbit Trail, and up to the El Casco locality were analyzed in the paleomagnetics laboratory at the University of Florida. Hehn et al. (1996) discussed in detail the methods, analysis, and results of the paleomagnetic sampling program conducted in the Eden Hot Springs and Jack Rabbit Trail sections. Other samples, including those stratigraphically above the El Casco locality and some resampled sites below were analyzed at the Berkeley Geochronology Center (BGC).

Samples were incrementally demagnetized and measured on a computerized 2-G Enterprises cryogenic magnetometer located in a magnetically shielded room. In Florida, demagnetization was accomplished using a combination of both thermal and alternating field techniques depending on the intensity of initial remanent magnetization (Hehn et al., 1996). At BGC, only thermal demagnetization techniques were performed. After measuring the intensity and direction of the natural remanent magnetization (NRM) of all samples, each was subsequently demagnetized by stepwise heating at 40 °C intervals starting at 120 and ending at 680 °C.

In order to determine the dominant magnetic mineral, Hehn et al. (1996) subjected 15 samples from the Eden Hot Springs–Jack Rabbit Trail section to isothermal remanent magnetization (IRM) analysis. Of these, 13 showed saturation magnetization behavior indicative of magnetite as the dominant carrier, whereas two samples showed behavior suggestive of hematite. Behavior during thermal demagnetization showed the removal of a low-temperature overprint component, possibly due to goethite, by 200 °C. At successively higher temperatures, consistent magnetization directions were generally obtained to about 560 °C, above which decreases in intensity and unpredictable changes in direction occurred. This supports the conclusion that magnetite was the primary carrier of the characteristic remanent magnetization (ChRM), given a Curie temperature for that mineral of about 580 °C. Additional steps to 680 °C were performed to help characterize components of magnetization, such as that due to hematite, that might overprint samples.

In some samples, demagnetization failed to result in a stable, linear vector decay toward the origin of the orthogonal vector diagrams on which the demagnetization data were plotted. Polarity could still be inferred, however, based on the position and trend of the horizontal and vertical components. Principal components analysis was performed on each sample to determine the least-squares best fit line of the characteristic component of magnetization (Kirschvink, 1980), and these data were then combined using Fisher statistics to determine site mean directions. Site classifications used in this report follow Johnson et al. (1982) rather than Opdyke et al. (1977), except in the case of class I sites, and differ somewhat from those used by Hehn et al. (1996). In this report, class I sites are those that have three statistically significant concordant directions. For three samples, a Fisher R value > 2.62 is statistically significant at the 95% confidence level. A class II site is one that is not statistically significant, but in which the polarity determination is not in doubt; i. e., usually a site in which two of three samples "have nearly parallel magnetic vectors with the third being divergent" (Johnson et al., 1982, p. 21). This can also include a site where one sample was lost or damaged but the remaining two have concordant directions. Class III sites were those having all three vectors widely divergent; these were rejected from the database. A class IV site is one in which only a single sample is available for analysis (as in Hehn et al., 1996). Class IV sites cannot be used alone for determination of the polarity of a particular section, but can corroborate polarity data from class I sites, thereby contributing qualitatively to the level of confidence with which one assigns the polarity of that section.

Of 198 sampled sites, four were lost, five were rejected on the basis of tenuous stratigraphic or structural associations, and six were rejected on the basis of class III status leaving 183 accepted sites. Of these, 141 were designated class I, 31 as class II, and 11 as class IV. As shown in Figure 5A and Table 1, the mean of all class I normal sites in both the Mount Eden and San Timoteo formations is D = 347.8°, I = 42.1° (k = 26.7, α_{95} = 3.2°, N = 75), and that of all class I reversed sites is D = 171.8°, I = –50.0° (k = 25.4, α_{95} = 3.5°, N = 66). The mean of all class I sites after inverting the reversed sites to the Northern Hemisphere is D = 349.5°, I = 45.8° (k = 24.6, α_{95} = 2.4°). Taking into account the ~10° average plunge (to the northwest) of the Badlands anticline, Hehn et al. (1996) calculated a declination anomaly, δ, of

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TABLE 1. FISHER STATISTICAL PARAMETERS FOR ALL CLASS 1 SITES

Section	N	R_1	k	α_{95}	Dec	Inc	ΔDec
All sites (N)							
Geo	75	73.3	37.2	2.7°	349.2°	53.2°	
Strat	75	72.2	26.7	3.2°	347.8°	42.1°	±4.3°
All sites (R)							
Geo	66	62.1	16.5	4.4°	173.4°	–61.8°	
Strat	66	63.4	25.4	3.5°	171.8°	–50.0°	±5.4°
Mount Eden Fm (N)							
Geo	15	14.6	33.3	6.6°	4.2°	51.0°	±10.5°
Strat	15	14.6	35.3	6.3°	350.3°	40.1°	±8.2°
Mount Eden Fm (R)							
Geo	18	17.0	17.5	8.5°	196.1°	–61.6°	±18.1°
Strat	18	17.3	25.3	7.0°	177.4°	–51.0°	±11.2°
ST–JRT (N)							
Geo	24	23.5	44.6	4.5°	348.7°	50.5°	±7.1°
Strat	24	22.9	21.0	6.6°	344.1°	41.0°	±8.7°
ST–JRT (R)							
Geo	29	27.5	18.7	6.4°	175.7°	–65.4°	±15.5°
Strat	29	27.8	23.9	5.6°	172.8°	–51.9°	±9.1°
ST–LEC (N)							
Geo	24	23.6	52.5	4.1°	345.4°	52.8°	±6.8°
Strat	24	23.4	36.6	5.0°	356.8°	41.2°	±6.6°
ST–LEC (R)							
Geo	17	16.4	26.7	7.0°	155.3°	–52.2°	±11.5°
Strat	17	16.4	28.8	6.8°	165.0°	–45.8°	±9.8°
ST–STLO (N)							
Geo	12	11.7	44.9	6.3°	336.5°	60.2°	±12.7°
Strat	12	11.7	46.6	6.1°	332.6°	46.9°	±8.9°
ST–STLO (R)							
Geo	2	1.9	165.1	13.8°	142.8°	–61.8°	±30.3°
Strat	2	1.9	162.0	13.9°	170.9°	–46.1°	±20.3°

Notes: N—number of samples averaged; R_1—Fisher statistic; k—precision parameter; Dec—declination; Inc—inclination; ΔD—declination confidence limit; ST–JRT refers to sites in the San Timoteo formation along Jack Rabbit Trail; ST–LEC refers to sites in the San Timoteo formation in the Landfill–El Casco section; and ST–STLO refers to sites in the San Timoteo formation in the San Timoteo Canyon–Live Oak Canyon section. (N) = normal sites; (R) = reversed sites.

–3.5°. Correcting for δ yields a mean direction of D = 353.0°, I = 45.8° (Table 2).

Because the sampling strategy involved two different formations of different lithology and depositional regime, and because the three major stratigraphic sections analyzed are somewhat separated geographically, a mean direction was also calculated for each of four separate groups of sites (Fig. 6 and Table 2). These groups are (A) the Mount Eden formation; (B) the San Timoteo formation along Jack Rabbit Trail; (C) the San Timoteo formation in the Landfill–El Casco composite section; (D) and the San Timoteo formation north of San Timoteo Canyon (the San Timoteo Canyon–Live Oak Canyon composite section).

To determine whether the normal and reversed populations are statistically antipodal at the 95% confidence level, i.e., whether the two populations share a common mean direction, a reversals test was conducted following criteria established in McFadden and McElhinny (1990). The calculated critical angle, γ_c, of 4.7° between the mean directions of the two populations indicates a positive reversals test of class A status (Fig. 5B). However, the calculated observed angle, γ_o, of 8.4° exceeds that of the critical angle, resulting in rejection at the 95% confidence level of the hypothesis of a common mean direction (McFadden and McElhinny, 1990, p. 727). In addition, fold tests, as well as differing mean inclinations for normal and reversed populations in groups A–C above, imply, as noted in Hehn et al. (1996), the possibility of the incomplete removal of a secondary field overprint. Such an overprint is also supported by the findings that, for the mean of all normal class I sites and for sites in groups B and C, the precision (or concentration) parameter, k, decreases in value, and the α_{95} circle of confidence increases, as the structural correction is applied to the geographic coordinates (i.e., the bedding is not horizontal); reversed sites, however, show an expected increase in clustering (Table 1). These data suggest the unlikely explanation that the normal polarity component was acquired

TABLE 2. FISHER STATISTICAL PARAMETERS PLUS ROTATION AND CONFIDENCE INTERVALS (REVERSED SITES ROTATED TO NORTHERN HEMISPHERE)

Section	N	R_1	k	α_{95}	Dec	Inc	ΔD	R_2	ΔR
All sites	141	135.3	24.6	2.4°	*353.0°	45.8°	±3.4°	–7.0°	±4.0°
Mount Eden fm[a]	33	31.8	26.3	5.0°	*357.3°	46.1°	±7.2°	–2.7°	±6.4°
ST–JRT[b]	53	49.9	16.8	4.8°	*352.6°	45.5°	±6.8°	–7.4°	±6.2°
ST–LEC[c]	41	39.7	30.2	4.1°	352.2°	43.2°	±5.6°	–7.8°	±5.3°
ST–STLO	14	13.7	46.3	5.7°	335.3°	47.0°	±8.3°	–24.7°	±7.2°

Notes: N—number of samples averaged; R_1—Fisher statistic; k—precision parameter; Dec—declination; Inc—inclination; ΔD—declination confidence limit; R_2—rotation; ΔR—rotation confidence limit; negative sign under R2 column denotes counterclockwise rotation; ST–JRT refers to sites in the San Timoteo formation along Jack Rabbit Trail; ST–LEC refers to sites in the San Timoteo formation in the Landfill–El Casco section; and ST–STLO refers to sites in the San Timoteo formation in the San Timoteo Canyon–Live Oak Canyon section. Mean of a + b + c = dec. 354.0° ± 5.4°, Inc. 45.0°, R ± ΔR = –6.0° ± 5.8° (α_{95} = 3.8°). Calculations based on Pliocene–Pleistocene (= current) reference pole for North America (λ = 90°N, ϕ = 360°, α_{95} = 3.0°) following Demarest (1983) and Butler (1992).

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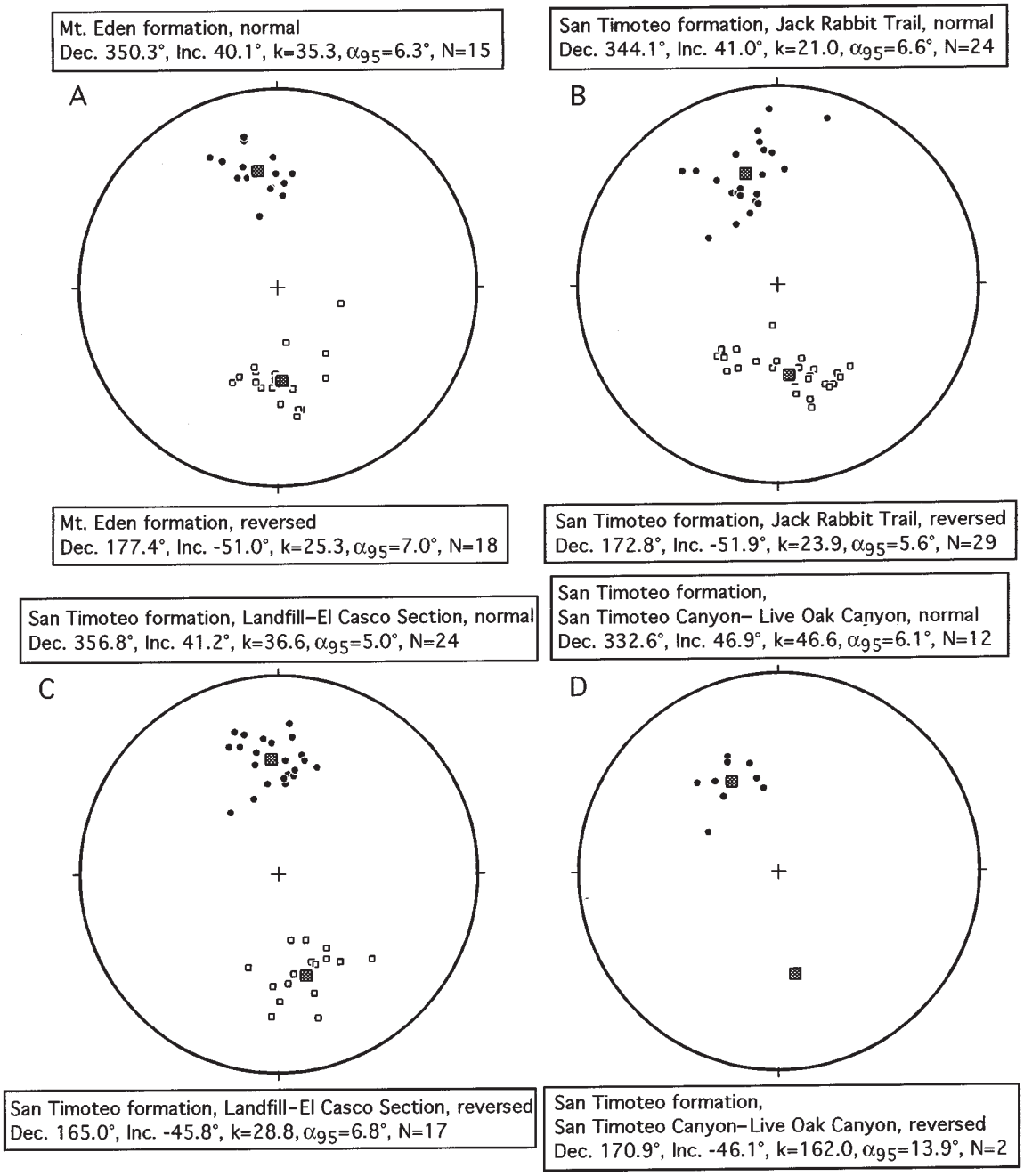


Figure 6. Equal-area stereo plots with Fisher statistical parameters for class I sites in (A) the Mount Eden formation, (B) the San Timoteo formation along Jack Rabbit Trail, (C) the San Timoteo formation in the Landfill–El Casco composite section, and (D) the San Timoteo formation in the San Timoteo Canyon–Live Oak Canyon composite section. Symbols as in Figure 5. Not corrected for declination anomaly due to plunging anticline. Note apparent counterclockwise deflection relative to Pliocene (and current) North American pole position.

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after folding, whereas the reversed polarity component was acquired prior to folding. A more reasonable explanation is the incomplete removal of a secondary field component resulting in the normal component being not fully isolated. It is with these caveats, therefore, that the revised magnetostratigraphy of the San Timoteo Badlands is presented.

Magnetic Polarity Zonation

Eden Hot Springs–Jack Rabbit Trail Composite Section. In the first phase of this project, Hehn et al. (1996, Figs. 6 and 7) developed a magnetostratigraphy for the lower part of the Badlands succession by combining four transects in the Eden Hot Springs area with one essentially continuous transect along Jack Rabbit Trail. In this report, the magnetostratigraphy of Hehn et al. is modified as a result of additional information obtained through more recent field work. Figures 7 and 8 show all Eden Hot Springs transects, the resulting composite section, the section along the lower Jack Rabbit Trail transect interpreted to represent the same interval, and the resulting correlation and overall composite magnetostratigraphy.

Hehn et al. (1996, Fig. 6) considered the reversed interval in which the Frick quarries occur the lowest reversal in the sequence, R1. Here, this reversed interval is designated R2 (Fig. 7) because a short normal-reversed-normal sequence (N1-R1-N2) was found stratigraphically below the Frick quarries level (but not at the same locality) (Fig. 8). The transect with N2 of Hehn et al. (their Mount Eden section 3) is also excluded from the composite section because more recent field work determined that its stratigraphic position is suspect due to faulting.

Transects 4, 5, and 6 are in the upper part of the Mount Eden formation in the Eden Hot Springs area (Mount Eden Section 1 in Hehn et al., 1996, Fig. 6) and are correlated on the basis of a thin (<5 cm) but distinctive whitish-green siltstone bed (Fig. 7). Near the base is a short normal polarity interval designated N3. Between N3 and the base of the ripple-laminated sandstone that caps transect 4 is a short reversed-normal-reversed interval (R3-N4-R4). This sequence is missing just below the ripple-laminated sandstone along Jack Rabbit Trail (Fig. 8). Thus it appears, as noted in Hehn et al. (1996), that the Mount Eden formation in the Eden Hot Springs section is more complete than along Jack Rabbit Trail, a consequence, perhaps, of the many faults in the area. The ripple-laminated sandstone is normally magnetized throughout its thickness along Jack Rabbit Trail and is designated N5.

Figure 9 shows the results for the remainder of the Jack Rabbit Trail transect, in the San Timoteo formation stratigraphically above the ripple-laminated sandstone, revealing four additional intervals of normal polarity (N6, N7, N8, N9) and five more intervals of reversed polarity (R5, R6, R7, R8, R9). The designated numbers of the paleomagnetic intervals differ slightly from those of Hehn et al. (1996) because an additional short interval of normal polarity is included in what was originally R5 based on results obtained from resampling sites 157–159. This does not, however, affect the correlation to the MPTS due to stratigraphic corrections made in lower parts of the section.

Riverside County Landfill–El Casco Composite Section. The Landfill–El Casco composite section consists of seven transects, the first three of which are located southeast of the landfill, the fourth within the landfill proper, and the remaining three from immediately north of the landfill to San Timoteo Canyon (Figs. 4B and 10). Because direct lithostratigraphic correlation with the Jack Rabbit Trail section is not possible, stratigraphically ascending polarity intervals in the Landfill–El Casco composite section are given a Greek letter suffix, rather than a number, following the "N" or "R" (e.g., Nα).

At the base of transect 1 is a reversed polarity interval about 60 m thick characterized by eight sites (205–211, 213), capped by normal polarity site 214. These intervals are designated Rα and Nα, respectively. A reversed polarity interval at the base of transect 2 and a short, ~15 m, normal polarity interval above are correlated to Rα and Nα on the basis of projections of strike and dip (Figs. 4B and 10). Nα is overlain by about 19 m of reversed polarity strata, Rβ, which, in turn, are overlain by another interval of normal polarity designated Nβ.

Transect 3, located about 0.4 km northwest of transect 2, comprises two superposed sections correlated on the basis of a marker conglomerate bed and includes about 85 m of normal polarity strata designated Nγ (sites 300–309; Figs. 4B and 10). Strike and dip projections do not preclude a correlation of Nγ to Nβ, but this cannot be confidently demonstrated in the field. Transect 4 is located about 0.7 km northwest of the third and was measured and sampled along the north wall of the landfill. It consists of about 70 m of normal polarity strata and, based on this, plus stratigraphic and field relations, likely overlaps somewhat with the previous transect. This interval, therefore, is also designated Nγ (Fig. 10). At the top of transect 4 is reversed polarity site 316, or Rγ.

North of the landfill begins the long, discontinuous stratigraphic section that includes transects 5–7, at the top of which is the El Casco locality on the south side of San Timoteo Canyon (Figs. 4B and 10). At the base of this section, beginning with site 317 in transect 5, is a reversed polarity interval considered a continuation of Rγ on the basis of bedding attitudes. About 7 m above site 317 is a normal polarity interval at least 30 m thick, Nδ, characterized by five sites, 317A–318B. Between Site 318B at the top of Nδ and Site 346, about 70 m below the El Casco locality, are nearly 400 m of section (transect 6; Fig. 10) where the magnetostratigraphy is poorly developed because of the paucity of fine-grained beds amenable to analysis. Of five widely distributed sites sampled in this interval, three show normal polarity (319, 320, 344A), whereas site 345, about 150 m below site 346, is of reversed polarity (site 344 was rejected). These intervals are designated Nε and Rε, respectively, because it cannot be demonstrated confidently that Nε is a continuation of Nδ (Fig. 10). Transect 7, within which all sites, 346–351, are of reversed polarity, is similarly labeled Rζ rather than a continuation of Rε. Site 350 is approximately equal in stratigraphic level to the original El Casco quarries.

San Timoteo Canyon–Live Oak Canyon Composite Section. Both sites of transect 1, 352–353, show reversed polarity. Although this interval may represent a continuation of Rζ, it is designated Rη for reasons similar to those given above (Fig. 11).

Transect 2 is stratigraphically above all those previously discussed, but an accurate measurement of the stratigraphic distance between transects 1 and 2 is not possible because of intervening facies changes and vegetation cover. Sites 354–359 are each spaced no more than 2 m apart and are all of normal polarity. This interval is designated Nη. Site 360, about 2 m above site 359, is of reversed polarity and designated Rθ.

Sites 400–406, collected in Live Oak Canyon from the stratigraphically highest transect, are all of normal polarity (Figs. 4C and 11). This interval is designated Nθ. "$Qvoa_1$" at the top of this transect is estimated to have an age of about 500 k.y. based on the stage of soil development (Kendrick, 1995, personal commun.; Kendrick et al., 1993; Kendrick and McFadden, 1996).

BIOCHRONOLOGY AND CORRELATION TO THE MAGNETIC POLARITY TIME SCALE

Temporal Duration

As an independent estimate of the temporal duration of the Jack Rabbit Trail stratigraphic section, Hehn et al. (1996) used the number of reversals

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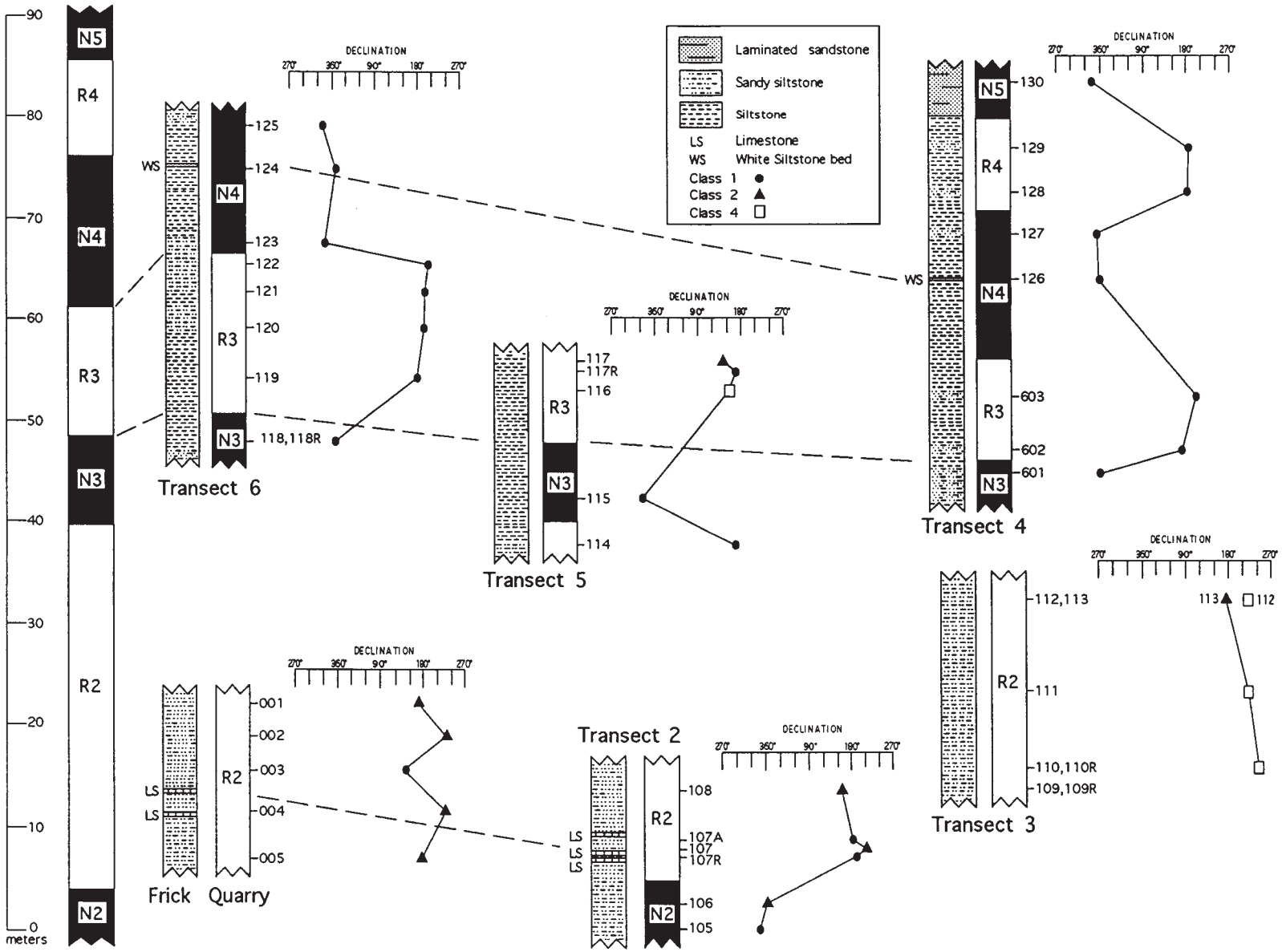


Figure 7. Lithostratigraphy, magnetostratigraphy, and correlation of sampled transects in the Eden Hot Springs area and resulting magnetostratigraphic composite. Lowermost normal polarity interval is designated N2 because an interval of normal polarity was found stratigraphically lower in the Jack Rabbit Trail section (see Fig. 8).

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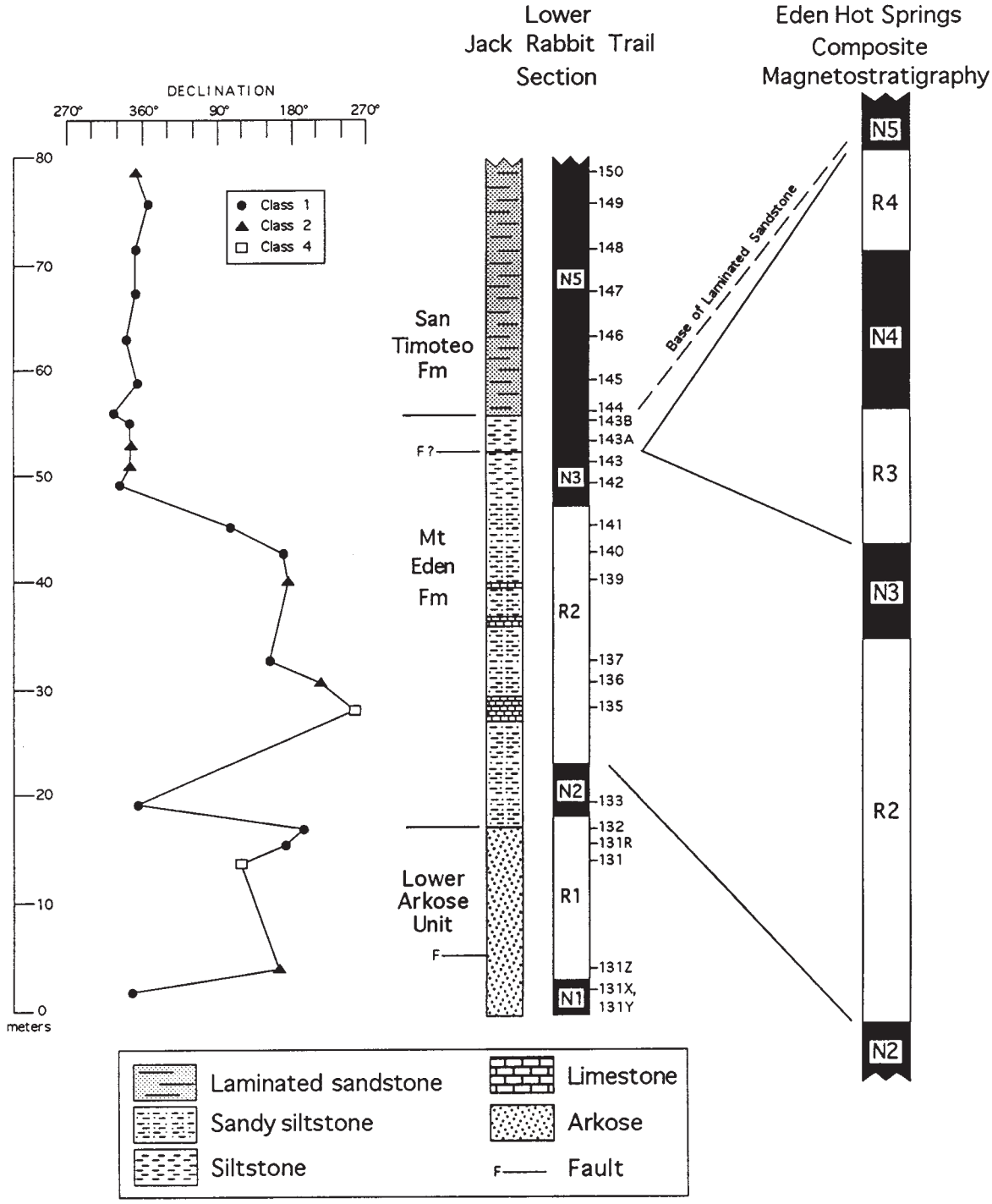


Figure 8. Lithostratigraphy and magnetostratigraphy of the lower Jack Rabbit Trail section with correlation to the Eden Hot Springs composite. Note the missing R3-N4-R4 sequence in the Jack Rabbit Trail section.

present to estimate the amount of time required for deposition (following Johnson and McGee, 1983). This tests the accuracy of the data having captured the number of reversals expected for the relevant section of strata. The equation from Johnson and McGee (1983) that relates the number of reversals and temporal duration is

$$\Delta t = S\tau N,$$

where Δt is the duration (in m.y.), $S = -\ln(1 - 2p)/2$ where $p = R/(N - 1)$, and R = number of reversals discovered by the sample set, τ = the mean time span of polarity intervals, and N = number of paleomagnetic sites. Hehn et al. (1996) calculated a duration of 4 m.y. for the Jack Rabbit Trail section, a duration consistent with biochronological criteria. Extending this procedure for the entire sequence, 183 sites (N) resolved 21 reversal events (R). There are also 21 reversal events between 0.5 and 5.5 Ma (the

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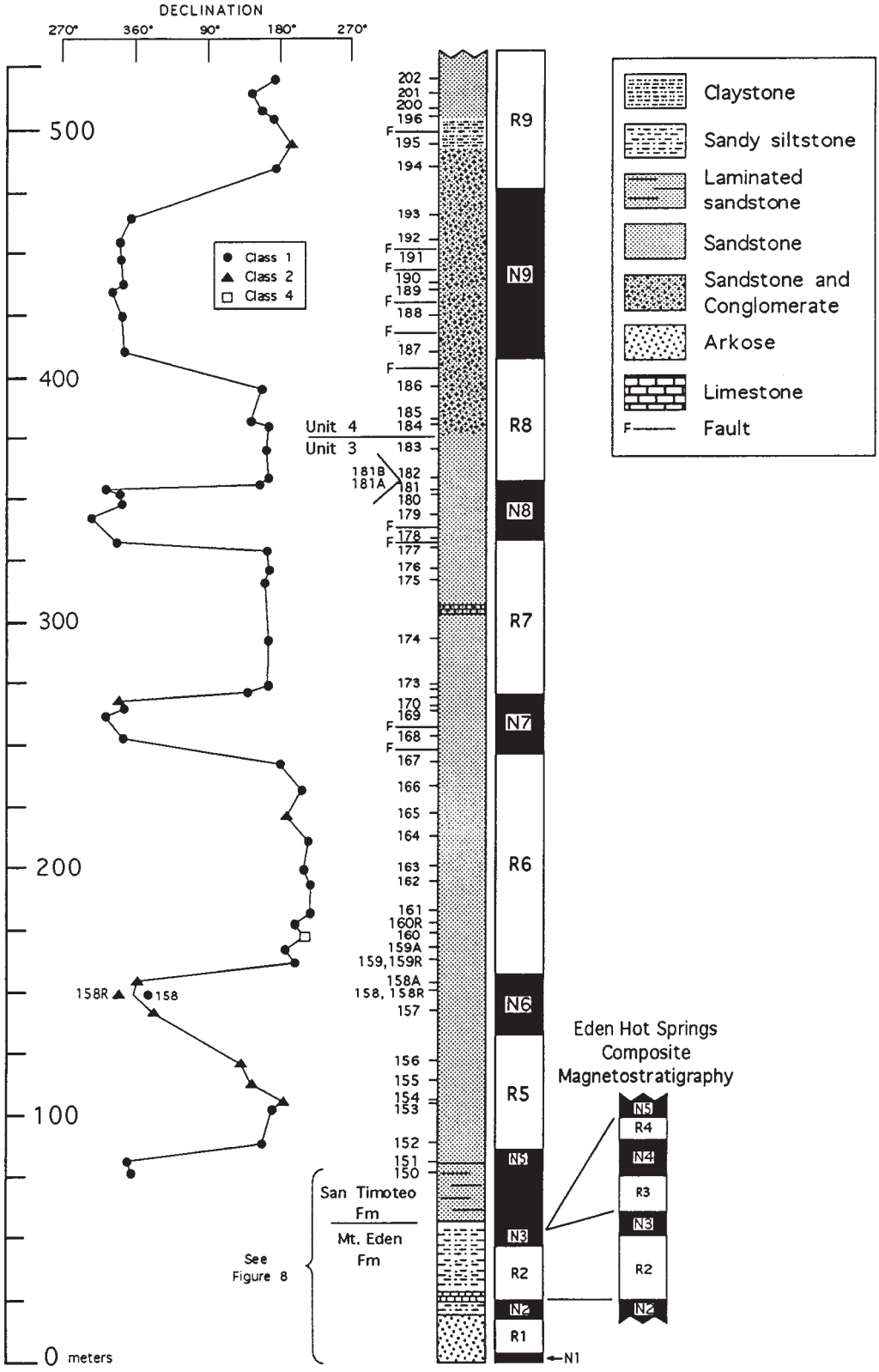


Figure 9. Lithostratigraphy and magnetostratigraphy of remaining Jack Rabbit Trail section.

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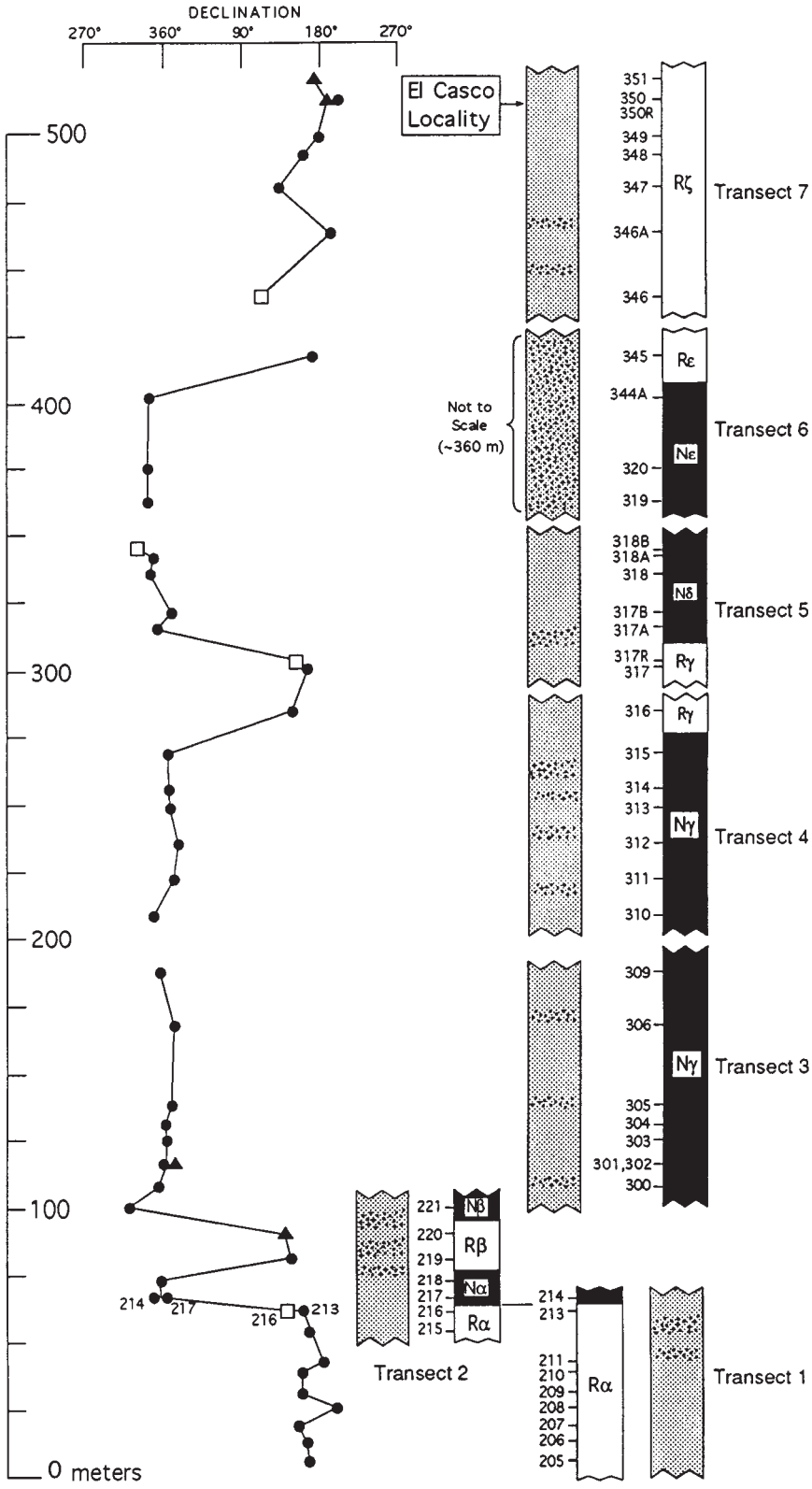


Figure 10. Lithostratigraphy and magnetostratigraphy of Riverside County Landfill–El Casco composite section. Refer to Figure 4B for spatial relationship of transects to one another. Overlap between transects 2, 3, 4, and 5 is likely (see text).

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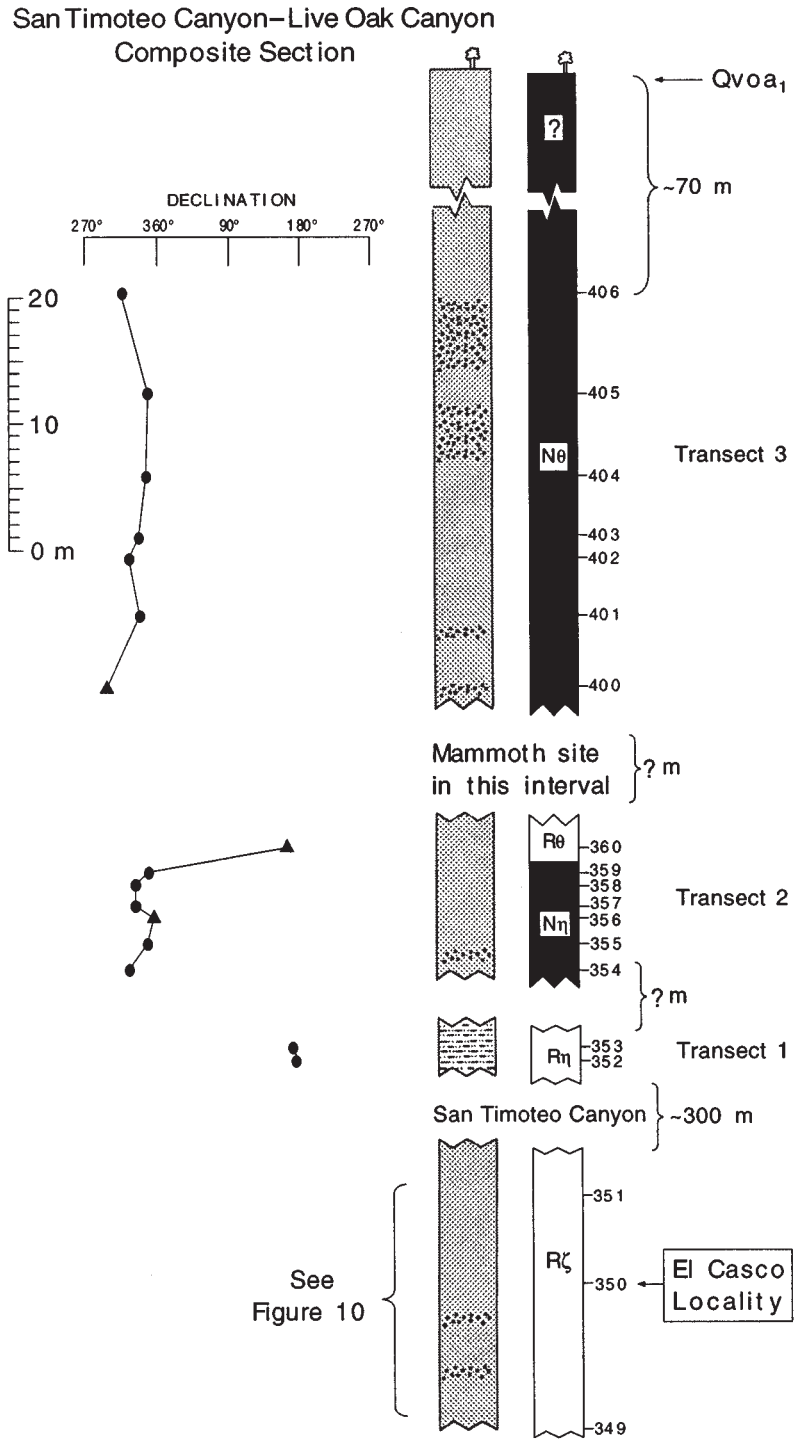


Figure 11. Lithostratigraphy and magnetostratigraphy of San Timoteo Canyon–Live Oak Canyon composite section. Mammoth site was not sampled for paleomagnetic analysis. Upper 70 m were not sampled due to lack of suitable outcrops. Note the change of scale from Figure 10.

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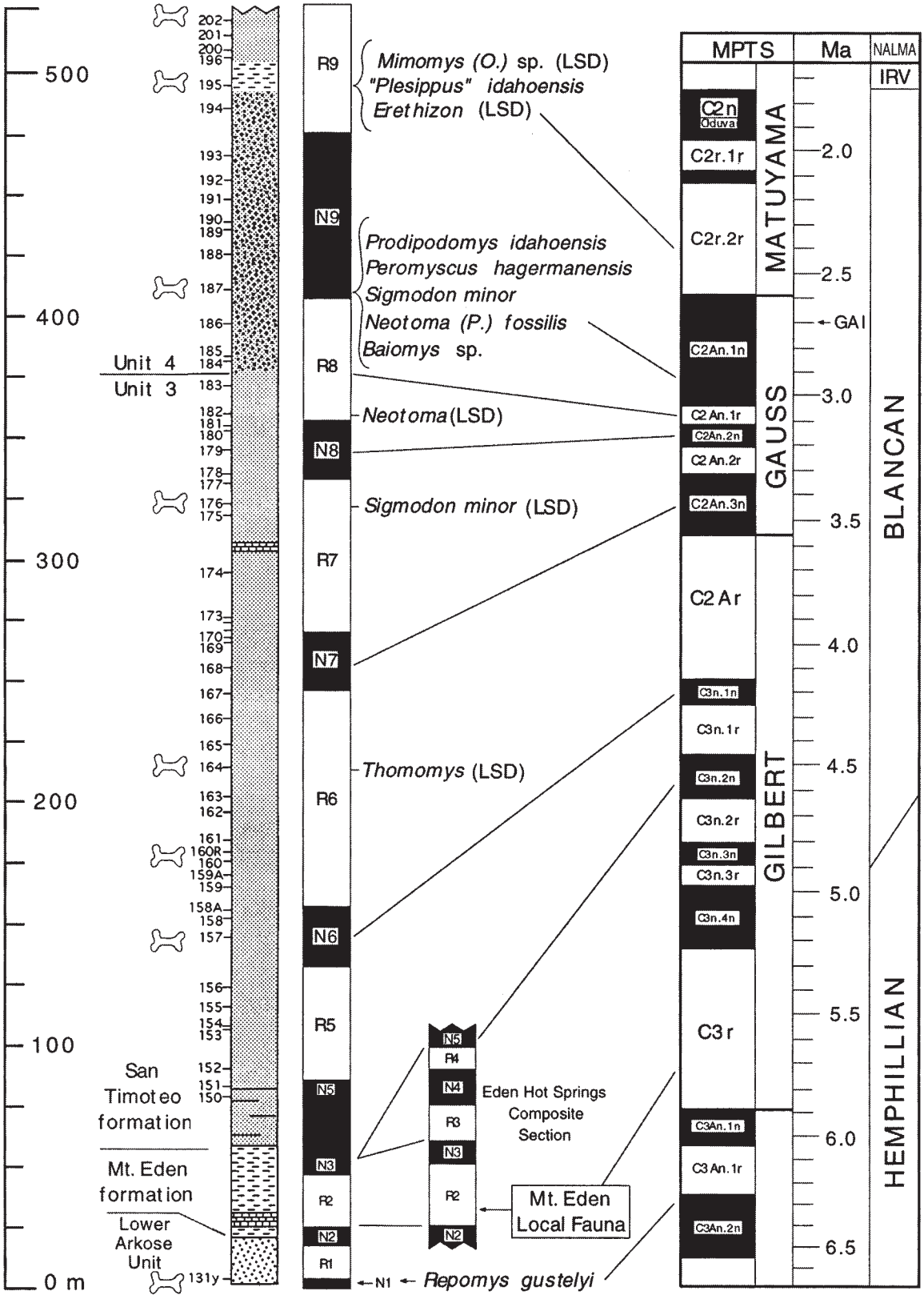


Figure 12. Biochronological calibration of the Eden Hot Springs–Jack Rabbit Trail composite section to the magnetic polarity time scale (MPTS). GAI—Great American Interchange. See text for discussion. LSD—Lowest stratigraphic datum; NALMA—North American Land Mammal "ages."

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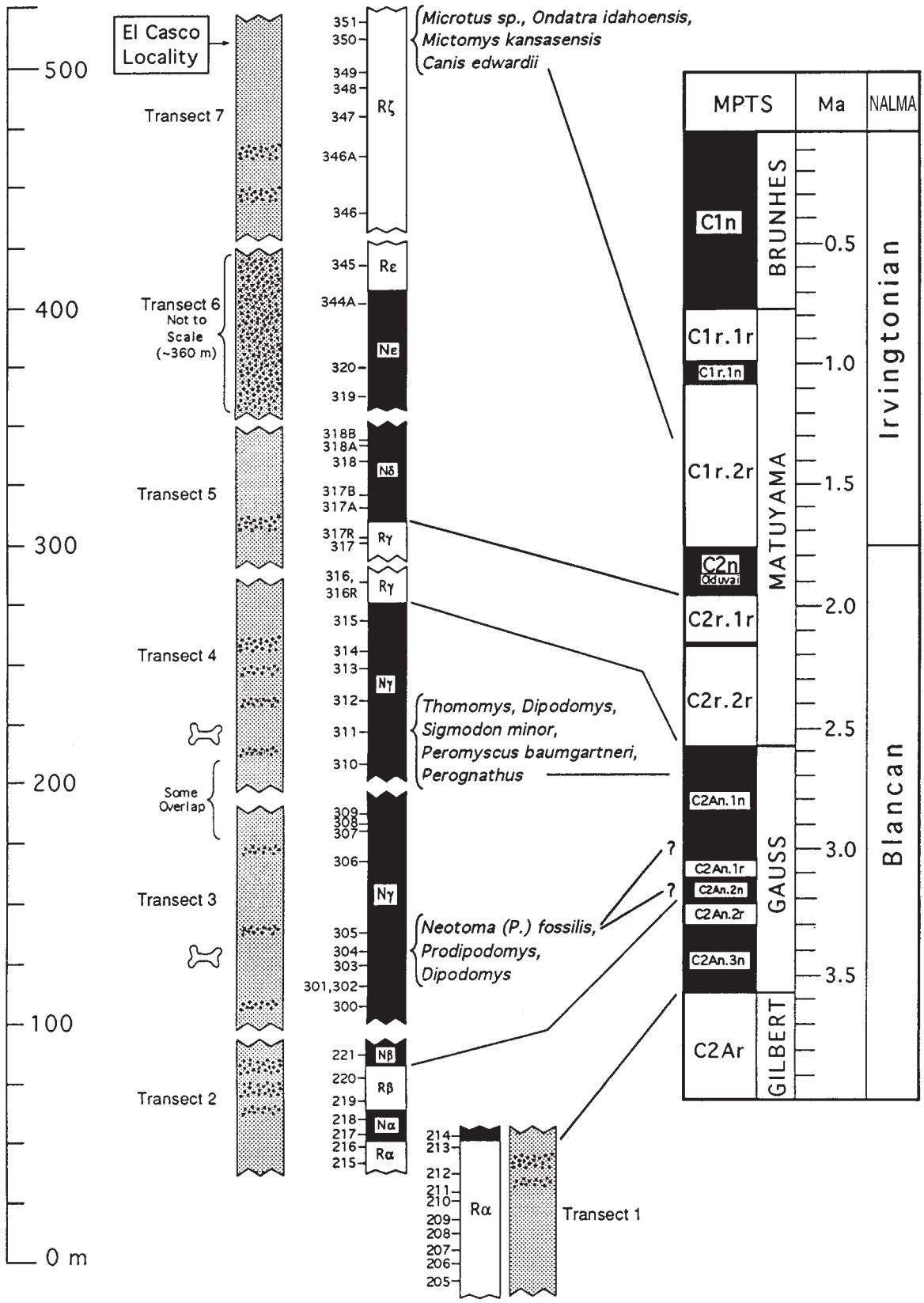


Figure 13. Biochronological calibration of the Riverside County Landfill–El Casco composite section to the magnetic polarity time scale (MPTS). NALMA—North American Land Mammal "ages." See text for discussion.

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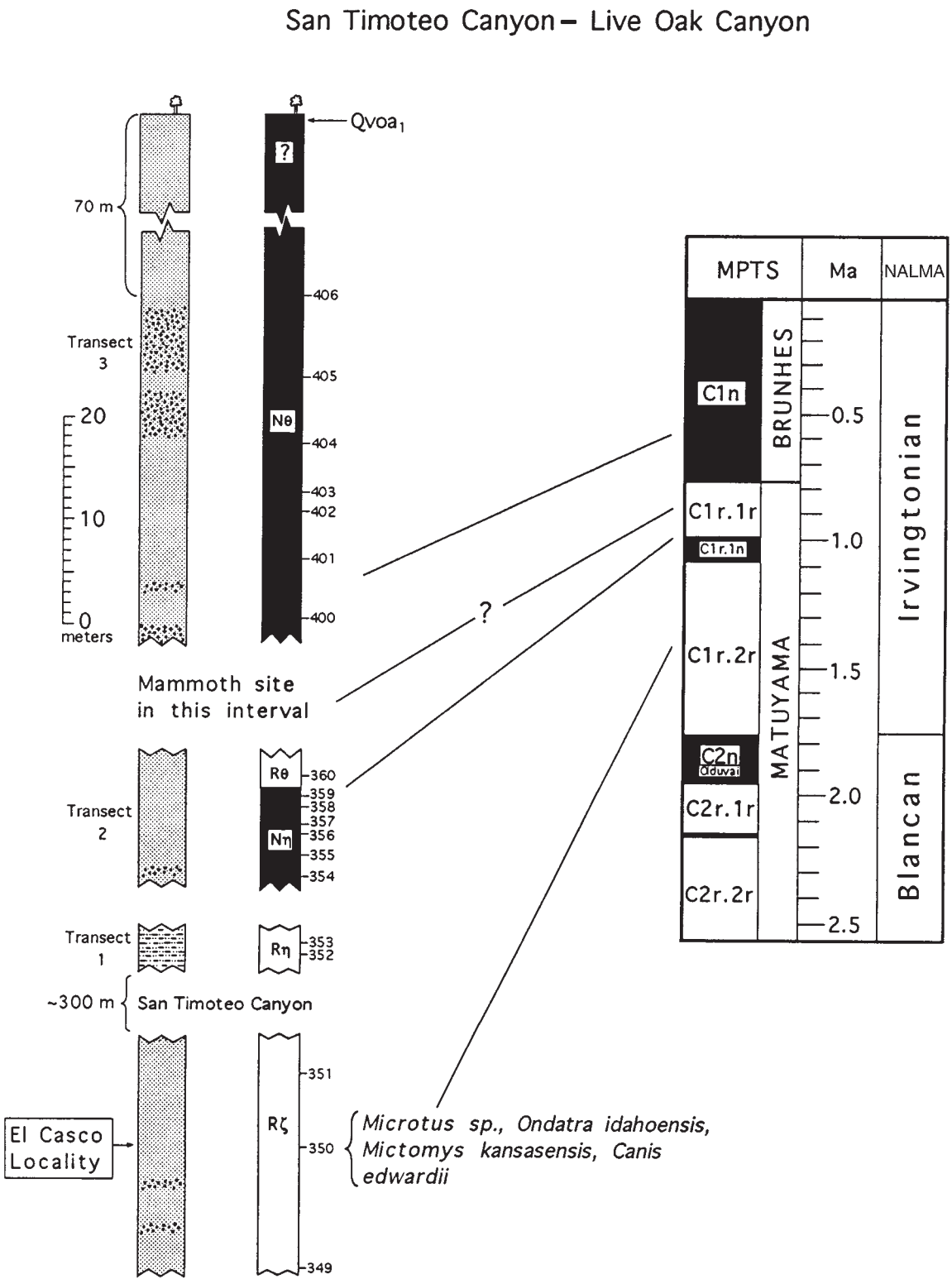


Figure 14. Biochronological calibration of the San Timoteo Canyon–Live Oak Canyon composite section to the magnetic polarity time scale (MPTS). NALMA—North American Land Mammal "ages." See text for discussion.

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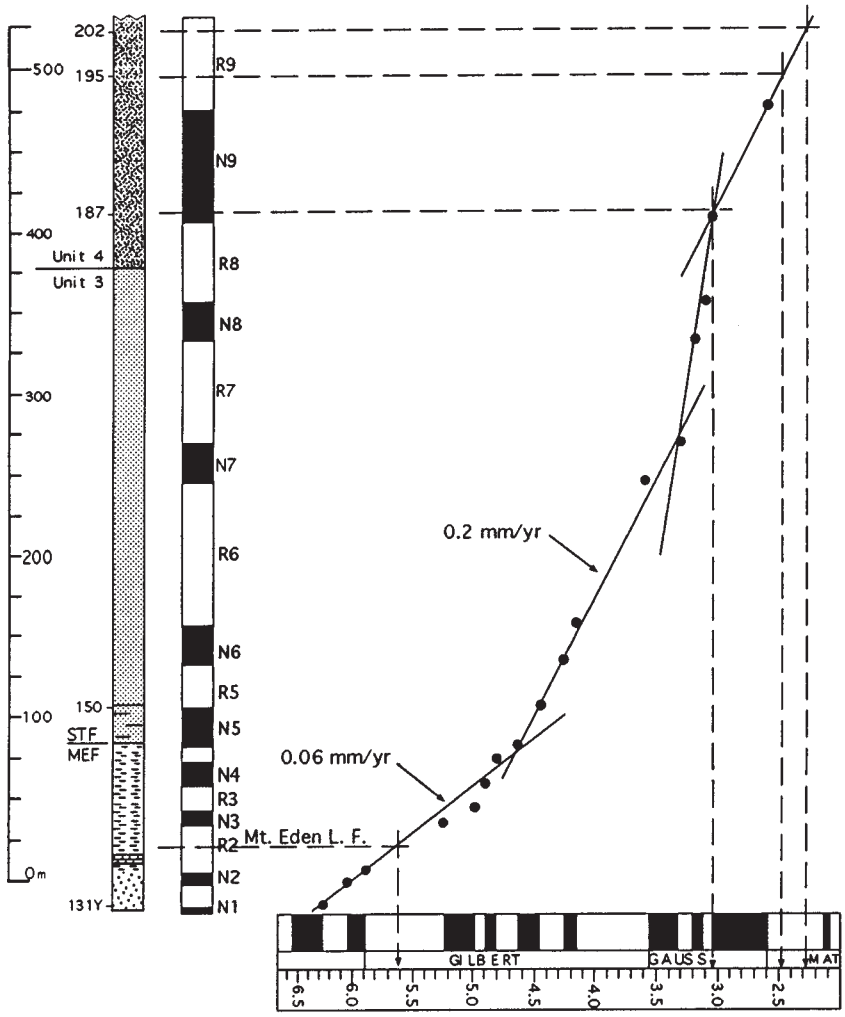


Figure 15. Correlation of the Eden Hot Springs–Jack Rabbit Trail composite section to the magnetic polarity time scale (MPTS) based on a plot of polarity reversal level as a function of age of reversal. Best fit lines result in a sediment accumulation history and also allow age extrapolation of significant fossil-bearing horizons. STF and MEF—San Timoteo formation and Mount Eden formation, respectively.

estimated ages, respectively, for the paleosol at the top of the section and the late Hemphillian Mount Eden local fauna near the base), according to the MPTS of Berggren et al. (1995), thus yielding a mean polarity interval duration (τ) of 0.24 m.y. The resulting duration expected for the entire sampled stratigraphic sequence is therefore 5.7 m.y., a reasonable estimate given the stratigraphic and structural complications.

Rate of Sediment Accumulation

Ages of events in a stratigraphic sequence are often estimated by extrapolation from the composite magnetostratigraphic section through a best-fit line to the MPTS, the line representing a mean sedimentation rate. However, the depositional environment of the Badlands sequence, particularly the San Timoteo formation, may compromise the ability to correlate the magnetostratigraphy to the MPTS on the basis of pattern recognition. Multiple unstable channels, high sediment influx over short time periods, and the ability to readily erode previously deposited material result in a stratigraphic section in which a considerable amount of time may be missing. Faults in the Badlands can compound the problem. Thus, polarity reversal boundaries in the section do not necessarily coincide with the actual reversal events in time. The truncation of a polarity reversal event will therefore affect the ability to interpolate the age of events located between magnetozone boundaries, as well as the accuracy of those age estimates. Sample spacing introduces additional uncertainty because reversal boundaries are typically placed midway between sites of opposing polarity.

Numerous papers have addressed these problems (e.g., May et al., 1985; Badgley et al., 1986; McRae, 1990), and techniques have also been developed based on the number and maturity of paleosols in stratigraphic sections to evaluate short-term sediment accumulation rates (e.g., Bown and Kraus, 1993; Kraus and Bown, 1993). Although paleosols are common in the Badlands sequence, the ability to count and laterally trace them as required by such techniques is impractical given the physical nature of the area (e.g., vegetation cover, discontinuous exposures, lack of laterally traceable marker horizons). Caution must be exercised, therefore, when extrapolating ages of events from the composite section through a best-fit line to the MPTS.

Nevertheless, the available biochronologic data from the Badlands combined with the relatively straightforward polarity reversal pattern of the Pliocene through Pleistocene epochs, as compared with, for example, the middle Miocene, limits the number of possible correlations to the MPTS.

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Figures 12–14 present the correlation of the magnetostratigraphy to the MPTS made on the basis of biochronology and magnetic reversal pattern.

Figure 15 correlates the stratigraphic level of magnetic reversals with the age of the reversals, resulting in best-fit lines that represent sediment accumulation history. For the fluvio-lacustrine Mount Eden formation, an accumulation rate of 0.06 mm/yr was calculated between the N2-R2 and R4-N5 boundaries (~80 m/1.3 m.y.). Woodburne et al. (1990) calculated similar rates (0.07 mm/yr) for lacustrine sections of the Barstow Formation, Mojave Desert. In the San Timoteo formation, fluvial deposition results in increased rates of accumulation with a calculated rate of 0.2 mm/yr (390m/1.6 m.y.).

Age estimation by extrapolation, discussed in the following, was only possible in the Eden Hot Springs–Jack Rabbit Trail composite (Figs. 12 and 15), where the 17 polarity reversals provide the points that can be plotted. For the Landfill–El Casco and San Timoteo Canyon–Live Oak Canyon sections, a similar plot cannot be constructed because no single transect, except transect 2 of the former section, has more than one polarity reversal and because accurate measurement of the stratigraphic distance between transects is hindered by problems noted previously (Figs. 13 and 14). However, it is still possible to estimate an approximate sediment accumulation rate for the Landfill–El Casco part of the section by considering the measured stratigraphic thickness between fossil sites 311 and 350. As detailed below, site 311 is correlated to the upper normal interval of the Gauss chron (C2n.1n), which brackets its age between 2.58 and 3.0 Ma. The age of site 350, the El Casco locality, is considered to be about 1.4 Ma. The ~500 m of section between these two sites results in a mean accumulation rate of 0.3–0.4 mm/yr. This rate is consistent with the lithology of this interval being representative of a higher energy environment of deposition than that lower in the San Timoteo formation.

Biochronology

Eden Hot Springs–Jack Rabbit Trail Composite Section. The rodent *Repomys gustelyi* was recovered from site 131y in the lower arkose unit stratigraphically below the mudstones from which the Mount Eden Local Fauna was obtained. Site 131y is in normal polarity interval N1 (Fig. 8). Originally described from the Warren Local Fauna, Mojave Desert (May, 1981). *R. gustelyi* is distinctly smaller than *R. maxumi* recorded from Mount Eden by May and Repenning (1982). May and Repenning (1982) considered the Warren Local Fauna older than the Mount Eden Local Fauna and placed the former near the C3An.1r–C3An.1n boundary. This supports correlation of site 131y to either chron C3An.1n or C3An.2n. Lithology similar to that containing the Mount Eden Local Fauna begins above this basal N1-R1-N2 sequence and includes an interval of reversed polarity (R2) correlated to chron C3r on the basis of the late Hemphillian age of the fauna. The presence of *Dinohippus osborni*, *Agriotherium gregoryi*, and *Teleoceras* sp. in the Mount Eden Local Fauna supports the late Hemphillian age, as these taxa are unknown in younger deposits (nor is *A. gregoryi* found in older deposits; May and Repenning, 1982; Tedford et al., 1987). The most reasonable correlation of the underlying N1-R1-N2 sequence, therefore, is with chron C3An, which refines correlation of the site 131y normal to C3An.2n (Fig. 12), or slightly older than the Warren Local Fauna. With an approximate age of 6.3 Ma for site 131y, the previously assumed late Miocene age for the lower arkose unit is confirmed. Consistent with the earlier correlation of Hehn et al. (1996), extrapolation from the Eden Hot Springs–Jack Rabbit Trail composite section to the MPTS yields an estimated age of about 5.6 Ma for the Mount Eden local fauna (Figs. 12, 15).

Fossils were first found stratigraphically above the Mount Eden formation in the lower San Timoteo formation between sites 160 and 165, within the relatively long interval of reversed polarity designated R6 (Fig. 12). The most common fossils recovered from these sites were those of the gopher *Thomomys* and a new species of primitive kangaroo rat. *Thomomys* is not known in deposits older than Blancan age. Correlation of the Mount Eden–Jack Rabbit Trail section to the MPTS places this interval in chron C2Ar.

Site 187, the next site in the sequence containing a significant rodent fauna, is located in the long interval of normal polarity designated N9. Taxa from this site include *Thomomys gidleyi*, *Prodipodomys idahoensis*, *Peromyscus hagermanensis*, *P. baumgartneri*, *Sigmodon minor*, and *Neotoma fossilis*, as well as a new species of *Baiomys* and *Peromyscus*, a second new species of kangaroo rat, plus a new genus of gopher. All are clearly indicative of the Blancan age and *Peromyscus hagermanensis* and *Neotoma fossilis* occur in medial Blancan aged faunas elsewhere (Tomida, 1987), thus supporting correlation of N9 to the Gauss normal chron (Fig. 12). Corroboration is provided by the presence of *Prodipodomys idahoensis*. The last recorded appearance (last appearance datum, LAD) of *P. idahoensis* is in the 111 Ranch and Wolf Ranch local faunas, Arizona (Harrison, 1978; Galusha et al., 1984; Tomida, 1987), near the Gauss-Matuyama boundary at 2.6 Ma. Additional support for correlation of site 187 to the upper part of the Gauss chron, or C2An.1n, is provided by the stratigraphic position of N9 below a reversed interval correlated to C2r.2r on the basis of microtine and hystricognathus rodents (see following). The stratigraphic position of site 187 in the lower part of the interval provides an estimated age of about 3.0 Ma (Figs. 12 and 15).

Sites 195 and 202A, near the top of the Jack Rabbit Trail section, also yielded diverse faunas. Three taxa in particular from site 195 provide the best biochronological calibration tie point of the local magnetostratigraphy to the MPTS, the microtine rodent *Mimomys* (*Ophiomys*) "*taylori-parvus*," the porcupine *Erethizon* ?*cascoensis*, and the horse *Plesippus idahoensis*. *Mimomys* (*O.*) "*taylori-parvus*" is a late Blancan species and its presence in an interval of reversed polarity (R9) above a normal interval correlated to the upper Gauss chron supports correlation of R9 to the reversal at the base of the Matuyama chron, or C2r.2r. The presence of *Erethizon* ?*cascoensis* strengthens this correlation because Neotropical immigrants are not known in U.S. faunas prior to the Great American Interchange ca. 2.7 Ma (Fig. 12). The first known record of *Erethizon* in the United States is in the Wolf Ranch Local Fauna (Harrison, 1978), which occurs near the Gauss-Matuyama boundary at 2.6 Ma. *Plesippus idahoensis* is known primarily from the late Blancan Jackass Butte Local Fauna of the Grand View sequence, Idaho, which is placed in chron C2r.2r on the basis of magnetobiostratigraphy (see Repenning et al., 1995, p. 60–61). The presence of these three late Blancan species in an interval of reversed polarity (R9) above an interval of normal polarity with medial Blancan indicators (N9) clearly supports correlation of this part of the section to chron C2r.2r. Figures 12 and 15 show the placement of site 195 at about 2.5 Ma.

Site 202A is ~20–30 m above site 195 and is the highest site sampled in the Eden Hot Springs–Jack Rabbit Trail composite section. In addition to *Mimomys* (*O.*) "*taylori-parvus*," both sites share *Sorex leahyi*, *Sylvilagus hibbardi*, *Thomomys gidleyi*, *Peromyscus hagermanensis*, *Sigmodon minor*, and *Neotoma fossilis*. In Figure 15, Site 202A is placed at about 2.3 Ma.

In summary, the Eden Hot Springs–Jack Rabbit Trail composite section is correlated to the MPTS primarily on the basis of (1) an interval at the base of the section with a short normal–reversed-normal sequence (N1-R1-N2) containing *Repomys gustelyi*; (2) a relatively long interval of reversed polarity (R2) in the Mount Eden formation containing the late Hemphillian Mount Eden Local Fauna; (3) a long interval of normal polarity (N9) in the San Timoteo formation with medial Blancan species; and (4) an interval of reversed polarity near the top of the section (R9) with diagnostic late Blancan taxa including *Mimomys* (*O.*) "*taylori-parvus*," the Neotropical immigrant *Erethizon*, and the horse *Plesippus idahoensis*.

Riverside County Landfill–El Casco Composite Section. The inability to accurately determine the amount of stratigraphic overlap, or missing

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section, between transects 2 and 3, 3 and 4, 5 and 6, and 6 and 7 precludes attempts to produce a best-fit line that would allow age estimation on the basis of extrapolation from a composite section to the MPTS, as was possible for the Eden Hot Springs–Jack Rabbit Trail section (Fig. 15). Correlation, therefore, was based solely on magnetic zonation pattern and biochronological considerations.

No fossils were collected from the first two transects of this composite section, but the presence of Blancan aged fossils in transects 3 and 4 and early Irvingtonian fossils from the El Casco locality near the top of the section (see below), together with the magnetostratigraphic pattern, support correlation to chrons C2Ar through C1r.2r with a duration of about 2 m.y. (Fig. 13). Implicit in this correlation is a degree of overlap with the Jack Rabbit Trail section greater than that assumed on the basis of lithostratigraphy alone.

Rodent fossils from site 304, about 20 m stratigraphically above the base of normally magnetized transect 3 (Nγ), include *Neotoma fossilis* and a species each of *Prodipodomys* and *Dipodomys*. The oldest record of *Dipodomys* is in the medial Blancan Taunton local fauna, Washington (Dalquest et al., 1992). Its occurrence in the Badlands sequence in a long interval of normal polarity supports correlation of Nγ to the Gauss chron. There is also a species of *Prodipodomys* from site 304 that appears to be the same as one from site 187 (not *P. idahoensis*, as site 187 includes more than one species). *Neotoma fossilis* from site 304 appears more primitive than that from site 187, suggesting that this part of Nγ correlates to an older level within the Gauss normal than the site 187 level.

If correlation of the reversed interval at the bottom of transect 1 (Rα) to chron C2Ar is correct, then the overlying short normal (Nα) and short reversal (Rβ) of transect 2 should represent chrons C2An.3n and C2An.2r, and the normal interval at the top of transect 2 (Nβ) should represent C2An.2n. The absence of a short interval of reversed polarity in transect 3 and all but the top of transect 4 implies that the short C2An.1r reversal (the Kaena subchron) is not recorded in the section or was missed in sampling (Fig. 13).

At the 15 m level of transect 4, site 311 produced several small mammal taxa including *Sylvilagus* sp., *Thomomys* sp., *Dipodomys* sp., *Peromyscus* ?*baumgartneri*, *Sigmodon minor*, two species of *Perognathus*, and the second species of new kangaroo rat. Stratigraphic considerations indicate that the lower part of this section and the upper part of transect 3 probably overlap, and the similar thicknesses of normally magnetized beds in both sections support this (Fig. 13). Both sections are considered part of the Gauss normal chron because long intervals of reversed polarity are not recorded. However, a reversed polarity site at the top of transect 4 (site 316, Rγ) suggests correlation of Nγ to the uppermost Gauss, or chron C2An.1n. Reversed interval Rγ, therefore, is correlated to chron C2r.2r. This correlation further implies that Nγ may be correlative with N9 in the Jack Rabbit Trail section and that sites 311 and 187 are nearly age equivalent.

In summary, stratigraphic and paleontological considerations indicate that transects 2–4 correlate primarily with the Gauss normal that extends from about 3.6 to 2.6 Ma. The base of transect 1 includes a relatively long interval of reversed polarity (Rα) that is correlated to chron C2Ar at the top of the Gilbert chron; there is no other long-term reversed interval in this part of the time scale. Reversed chron C2r, between the top of the Gauss normal and base of the Olduvai normal (chron C2n), spans ~750 k.y., but C2n is not long enough to account for the thickness of the normal sections discussed here without invoking extraordinarily high rates of deposition (85 m in 180 k.y., or 0.47 m/k.y.). Therefore, the most reasonable correlation of this part of the section, as shown in Figure 13, is from chron C2Ar at the top of the Gilbert chron, through the Gauss normal, C2An, to the reversal at the base of the Matuyama chron, C2r.2r.

Reversed polarity site 317 at the base of transect 5 immediately north of the landfill likely represents a continuation of Rγ, or chron C2r.2r (Fig. 13). The overlying normal (Nδ), therefore, may represent either C2r.1n (the Reunion normal) or C2n (the Olduvai). Because C2r.1n has a duration of only about 10 k.y., and Nδ is at least 35 m thick, correlation of the latter to C2n is most reasonable.

In the magnetically poorly characterized transect 6 between sites 318B and 346, no fossils are known. Site 350, however, near the top of transect 7 (Rζ), is nearly equivalent in stratigraphic level to the original El Casco quarries. Biochronological evidence discussed in the following supports placement of the El Casco locality in reversed chron C1r.2r above the Olduvai normal.

Three different microtine rodents are known from El Casco: the vole *Microtus* sp., the muskrat *Ondatra idahoensis*, and the bog lemming *Mictomys kansasensis*. The oldest occurrence of *Microtus* (sensu Repenning, 1992) anywhere in the United States is in the Vallecito Creek sequence, Anza-Borrego Desert (Zakrzewski, 1972), at a level Repenning (1992, p. 59, 78) and Repenning et al. (1995, p. 29) considered to be about 1.4–1.6 Ma. This consideration together with the primitive morphology of the El Casco species of *Microtus* implies the unlikelihood that the El Casco reversed interval is below the Olduvai normal; this would place *Microtus* there >2 m.y. ago. Further corroboration is provided by the *Ondatra idahoensis* datum established by Johnson et al. (1975) and Lindsay et al. (1990) on the basis of the presence of this taxon in the California Wash local fauna, Arizona. The California Wash local fauna is above the 2.0 Ma Huckleberry Ridge ash and below the base of the Olduvai normal (1.95 Ma). The occurrence, therefore, of *Ondatra idahoensis* in a long interval of reversed polarity also supports placement of the El Casco site above the Olduvai subchron in C1r.2r.

The El Casco bog lemming, *Mictomys kansasensis*, has been found associated with *Ondatra annectens*, a more advanced muskrat than *O. idahoensis*, in, for example, the early Irvingtonian Java local fauna (Martin, 1989). However, it is important to note that the muskrat at El Casco is an advanced form of *O. idahoensis*, whereas the bog lemming there is a primitive form of *M. kansasensis*. The El Casco local fauna appears to include both taxa at the opposite extremes of their respective temporal ranges. Furthermore, *Mictomys anzaensis*, from the same level in the Vallecito Creek succession as the species of *Microtus* noted here (Repenning et al., 1995), is considerably more primitive than the lemming at El Casco. The El Casco species, in turn, is more primitive than 1.2 m.y. old specimens of *M. kansasensis* from the Great Plains. Together, these biochronological considerations imply an age for the El Casco local fauna of between about 1.4 and 1.2 Ma, restricting placement of the El Casco locality to reversed interval C1r.2r (Fig. 13).

San Timoteo Canyon–Live Oak Canyon Composite Section. Fossils were not recovered from the three transects sampled for paleomagnetic analysis on the north side of San Timoteo Canyon, although a tooth of *Mammuthus* was recovered from a locality stratigraphically between transects 2 and 3 (Figs. 4C and 14). Nevertheless, a reasonable correlation of this composite section to the MPTS is still attainable on the basis of information provided here with respect to correlation of the El Casco locality, as well as on the presence of *Mammuthus*.

Mammuthus characterizes the Irvingtonian North American Land Mammal "age" and its first record in North America coincides with chron C2n, the Olduvai subchron (Lundelius et al., 1987), placed between 1.77 and 1.95 Ma by Berggren et al. (1995). It is interesting that *Mammuthus* has not been recorded from the San Pedro Valley, Arizona, or from the Vallecito Creek sequence in the Anza-Borrego Desert, although it is known from reversed polarity beds of the Ocotillo Formation, Borrego Badlands, below the Jaramillo normal subchron, C1r.1n (Remeika and Jefferson, 1993; Jefferson and Remeika, 1994). The age of the Borrego Badlands specimen, therefore, is bracketed between about 1.07 and 1.77 Ma. *Mammuthus* is also known from the Irvington local fauna in northern California (Savage, 1951) which Lindsay et al. (1975) placed in the upper

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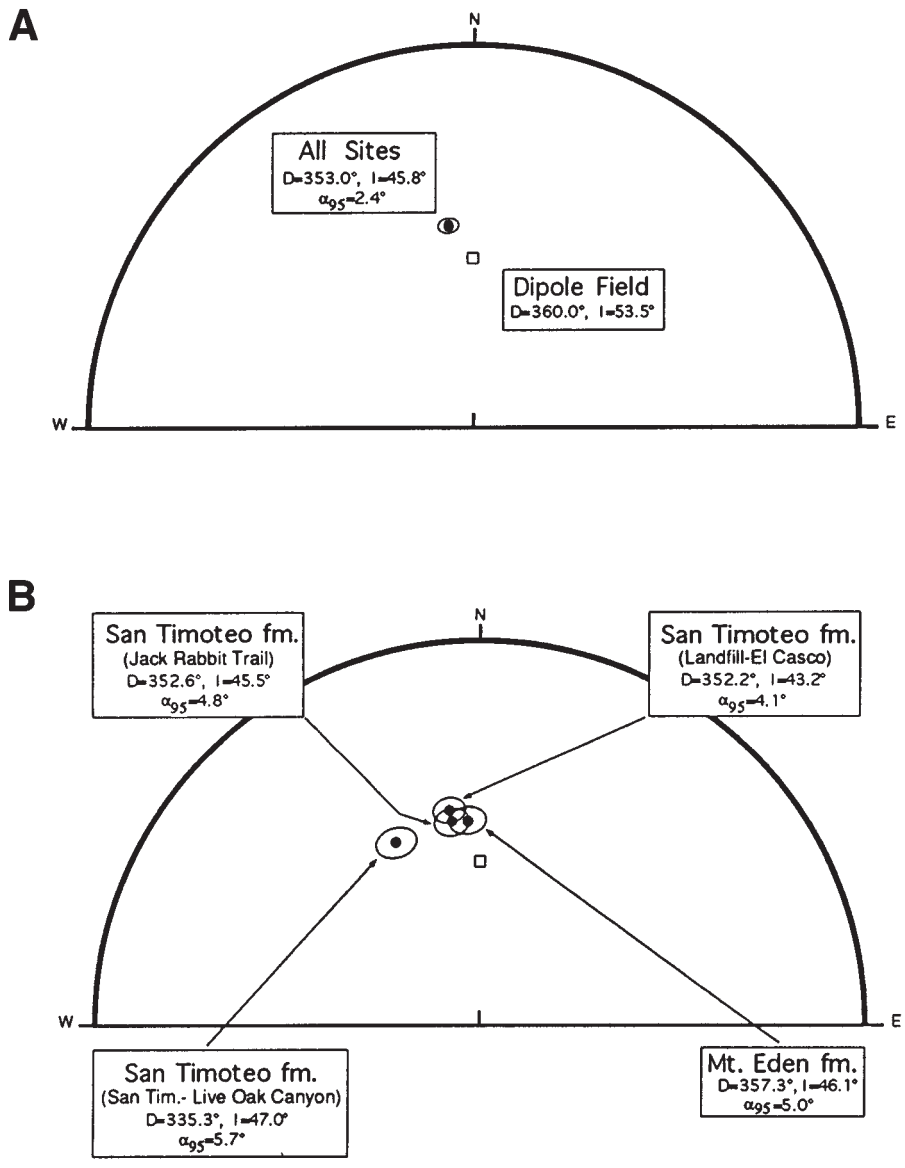


Figure 16. (A) Mean direction for all class I sites (reversed sites rotated to Northern Hemisphere; corrected for declination anomaly due to plunging anticline). (B) Mean direction for all class I sites in groups A, B, C, and D of Figure 6. Note anomalous direction of the mean of group D relative to groups A, B, and C. Square—present North American pole position.

Matuyama chron, noting that the fossils were collected from reversely magnetized strata.

Although the two reversed polarity sites, 352–353, located across San Timoteo Canyon from the El Casco locality are stratigraphically above the latter, it cannot be determined whether they represent the same reversed interval. However, because sites 354–360 in transect 2 are stratigraphically above both the El Casco locality and sites 352–353, the normal to reverse change (Nη to Rθ) in transect 2 is likely that from the top of the Jaramillo normal (C1r.1n) to the reversed interval (C1r.1r) that precedes the current Brunhes normal (Fig. 14). The stratigraphic position of the *Mammuthus* locality above site 360 implies correlation to chron C1r.1r, or between about 0.78 and 0.99 Ma. Based on correlations here, the long interval of normal polarity in the stratigraphically highest transect at Live Oak Canyon can only be the current Brunhes normal.

It is important to note one additional site on the north side of San Timoteo Canyon that was sampled for magnetic polarity. About 4.4 km northeast of the El Casco locality is the Shutt Ranch fossil locality (Fig. 1B; north-central part of S15, T2S, R2W, El Casco USGS 7.5′ quad). Located just below a regionally extensive geomorphic surface ("Qof" of Matti et al., 1992) with soil profiles less mature than those of "$Qvoa_1$" at the top of the Live Oak Canyon section, Shutt Ranch is stratigraphically well above site 360. Furthermore, species of *Microtus* and *Neotoma* from there are much more advanced than species of the same genera from the El Casco locality. Thermal demagnetization of samples from Shutt Ranch revealed declinations and inclinations indicative of normal polarity but with a demagnetization trend indicating the possibility of reversed polarity with a strong normal overprint. Together, the paleontological, stratigraphic, and magnetic data indicate an age no older than 0.99 Ma and possibly younger than 0.78 Ma, sig-

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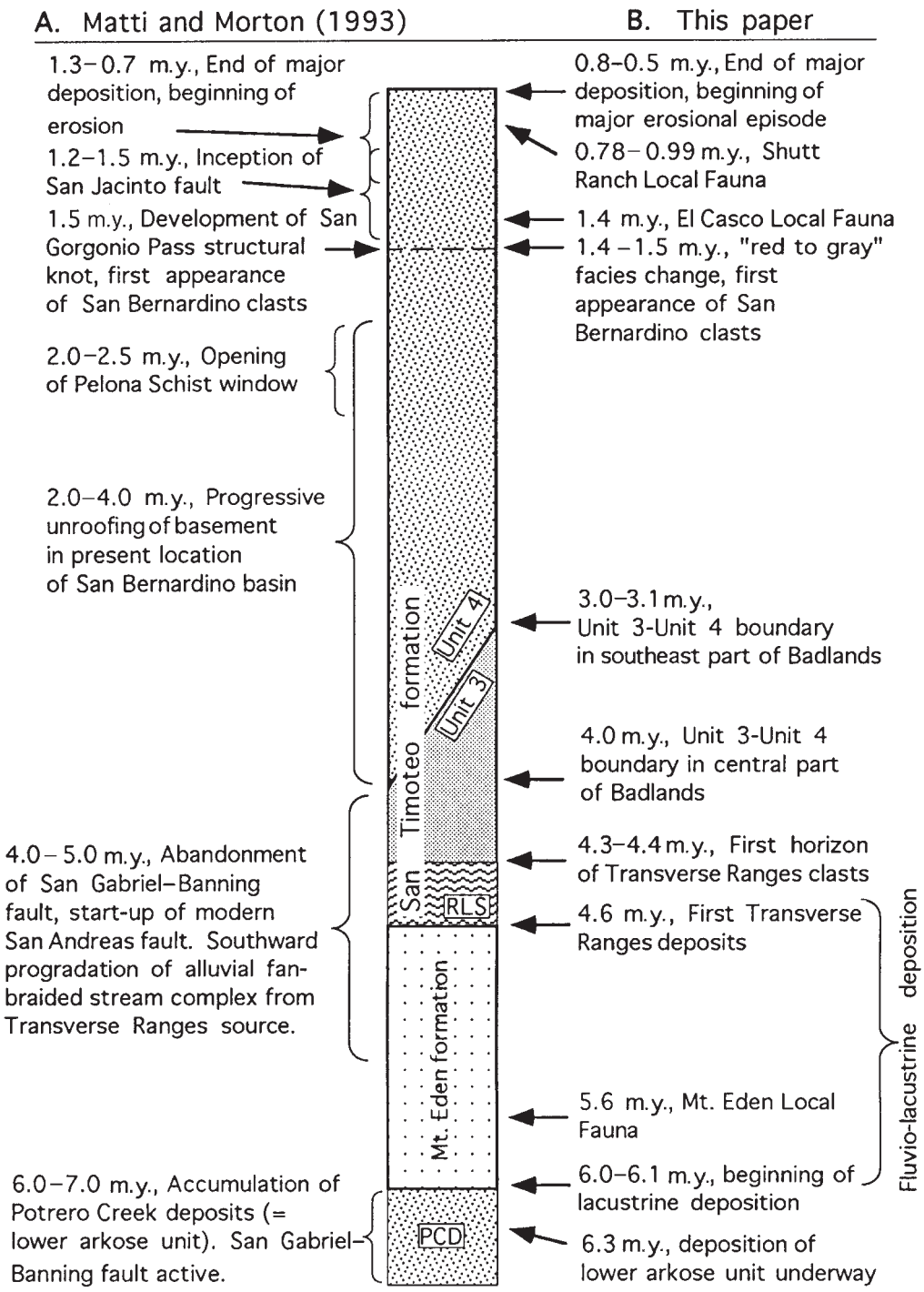


Figure 17. The Badlands sequence with the chronology of regional and local events placed at their respective stratigraphic and chronologic positions. Column A provides the generalized chronology of the regional events as discussed by Matti and Morton (1993); column B is a refinement of that chronology based on data assembled in this study. Diagonal line forming boundary between units 3 and 4 of the San Timoteo formation represents the earlier appearance of unit 4 in the northwestern region of the Badlands than in the southeastern. PCD—Potrero Creek deposits (including lower arkose unit); RLS—Ripple-laminated sandstone member of the San Timoteo formation.

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nificantly younger than Repenning's (1987, Fig. 8.1) placement of this fauna in the Olduvai normal (1.77–1.95 Ma).

TECTONIC IMPLICATIONS

Vertical-Axis Rotation of the Badlands

Previous seismic, geodetic, and paleomagnetic studies addressing block rotation in the San Jacinto fault zone (e.g., Johnson et al., 1983; Bogen and Seeber, 1986a, 1986b; Nicholson et al., 1986; Scheuing et al., 1988, 1990) concluded that significant (~20°–35°) clockwise vertical-axis rotation occurred over the past 1.3 m.y. on a series of small rigid blocks as a result of slip along secondary left-lateral conjugate faults in a regional field of right-lateral shear (for an alternate interpretation see Jones, 1988). Similar studies of a more regional nature (e.g., Luyendyk et al., 1985) also indicate large clockwise vertical-axis rotations in Neogene rocks of southern California.

In contrast, Hehn et al. (1996) provided evidence to infer 20.7° ± 5.2° of counterclockwise rotation based on a mean declination of 339.3° ± 4.2° for class I sites in the San Timoteo formation along the upper part of the Jack Rabbit Trail section (sites 170 to 202). New results incorporating additional sampled localities prompt reexamination of this apparently anomalous rotational direction.

Table 2 and Figure 16A show 7.0° ± 4.0° of counterclockwise vertical-axis rotation relative to the Pliocene (and current) North American dipole field direction (D = 360°, I 53.5° expected for site λ^s = 34°N, ϕ^s = 243°E calculated from the Pliocene [and current] North American reference pole: λ^p = 90°, ϕ^p = 360°, A_{95} = 3.0) based on a mean declination for all class I sites of 353.0° ± 3.4° (reversed sites rotated to Northern Hemisphere; confidence intervals for declination and rotation calculated following Demarest, 1983; Butler, 1992). Each of the four separate groups for which site mean directions were calculated also show counterclockwise rotation (Fig. 6; Table 2). However, when the normal and reversed polarity data are analyzed separately for each group, sites of normal polarity appear to be deflected counterclockwise to a greater degree than sites of reversed polarity, although in the Landfill–El Casco section the opposite is observed. This corroborates the problem noted previously regarding the apparent incomplete removal of an overprint component of magnetization and the inability to completely isolate the characteristic remanent magnetization for sites of normal polarity.

To reduce this overprint bias, reversed polarity sites were rotated to the Northern Hemisphere in order to allow the calculation of a single mean for all sites in each of the separate groups (Table 2, Fig. 16B). Because the plunge of the anticline in the Badlands shallows to the northwest, the declination anomaly correction of –3.5° was applied only to the mean direction of the Mount Eden formation and to the Jack Rabbit Trail section of the San Timoteo formation. For the Mount Eden formation, the declination and rotation confidence limits indicate negligible vertical axis rotation (Table 2). Hehn et al. reached a similar conclusion. Note also in Figure 16B that A_{95} of the mean direction of sites in the San Timoteo formation, exclusive of the San Timoteo Canyon–Live Oak Canyon section, overlaps with that of the Mount Eden formation. In contrast, the San Timoteo Canyon–Live Oak Canyon section has a mean declination of 335.3° ± 8.3°, which suggests significant counterclockwise rotation of –24.7° ± 7.2°. One explanation for this high degree of rotation may lie in the reduced number of sites available for calculating a mean (N = 14). Furthermore, only two of the 14 class I sites in this section are of reversed polarity, possibly increasing any bias that may have been imposed by an overprint.

Averaging together all class I sites in the Mount Eden and San Timoteo formations (reversed sites inverted), again excluding those in the San Timoteo Canyon–Live Oak Canyon section, yields a mean direction of 354.0° ± 5.4° (I = 45.0°) which results in 6.0° ± 5.2° of counterclockwise rotation. Although the data indicate that there may be a slight degree of counterclockwise vertical-axis rotation in the Badlands, a significant portion appears to be the result of an unremovable secondary magnetization. Another contributing factor may be the variation in plunge of the Badlands anticline, which shallows to the northwest.

Nevertheless, if the counterclockwise vertical-axis rotation in the Badlands region is real, then an explanation is warranted. Two are proposed: first, counterclockwise rotation could occur as a result of the northward translation of the San Jacinto Mountains block into the left bend of the San Andreas fault that was developing in the San Gorgonio Pass area about 1.5 m.y. ago (Matti and Morton, 1993). Second, there is also a left-stepping compressional bend in the San Jacinto fault just south of the San Bernardino basin and immediately adjacent to the northwestern part of the Badlands that may have contributed, particularly considering the proximity of this left step to the section that shows the greatest degree of apparent counterclockwise rotation, the San Timoteo Canyon–Live Oak Canyon section (Fig. 1). It is this local feature to which Morton and Matti (1993b) attributed uplift of the Badlands. It is important to note, however, that this area is highly complicated and not fully understood in a structural sense. Only further field work and more detailed geophysical and geological analyses, including additional paleomagnetic studies both within and outside the area bounded by the San Andreas and San Jacinto fault zones, as well as a consideration of slip-rate partitioning between these two systems, will resolve more accurately the nature of the apparent counterclockwise vertical-axis rotation in the Badlands area.

Regional Tectonic and Depositional Patterns

In their discussion on the paleogeographic evolution of the San Andreas fault zone in southern California, Matti and Morton (1993) and Morton and Matti (1993a) presented a general chronology for depositional and tectonic events in the region, including evolution of the Badlands. Figure 17A summarizes this generalized chronology. Figure 17B adds refinements based on new data discussed here and identifies certain events of a more local nature not addressed in previous studies. The following discussion traces the history of the Badlands region from the oldest events recorded in the sequence stratigraphically upsection to the most recent.

Matti and Morton (1993) proposed that deposition of the Potrero Creek beds, which include the lower arkose unit at the base of sequence in the study area, began about 6.0–7.0 Ma, based in part on the late Hemphillian age of the Mount Eden local fauna in overlying strata (Fig. 17A). Although onset of deposition of the lower arkose unit cannot be determined from information obtained in this study, deposition was in progress by at least 6.3 Ma and continued until about 5.9–6.0 Ma (Fig. 17B). The arkose was being shed generally to the north and west from eroding Peninsular Ranges basement highs located south and east of the San Timoteo basin (Fig. 18A).

Around 6.0 Ma, deposition of the fluvio-lacustrine mudstones that characterize the Mount Eden formation began (Fig. 17B), perhaps reflecting a lowering in base level associated with subsidence of the Salton Trough in response to late Miocene activity on the nearby San Gabriel–Banning fault. The Mount Eden local fauna, with an age of 5.6 Ma, was recovered from the lower part of the lacustrine beds. The depositional environment represented by the Mount Eden formation persisted until about 4.6 Ma (Fig. 17B). At that time, the San Timoteo basin began to fill with somewhat coarser-grained sediments exemplified by the ripple-laminated sandstone member at the base of the San Timoteo formation. The appearance in this unit of rare clasts of San Gabriel–type basement (Matti, 1996, personal commun.) signals a major shift in provenance for material being deposited into the Badlands basin from the previous Peninsular Ranges source. Not until about 4.3 Ma, however, low in unit 3, is there a distinct bed of San Gabriel–type clasts (Figs. 17B and 18B).

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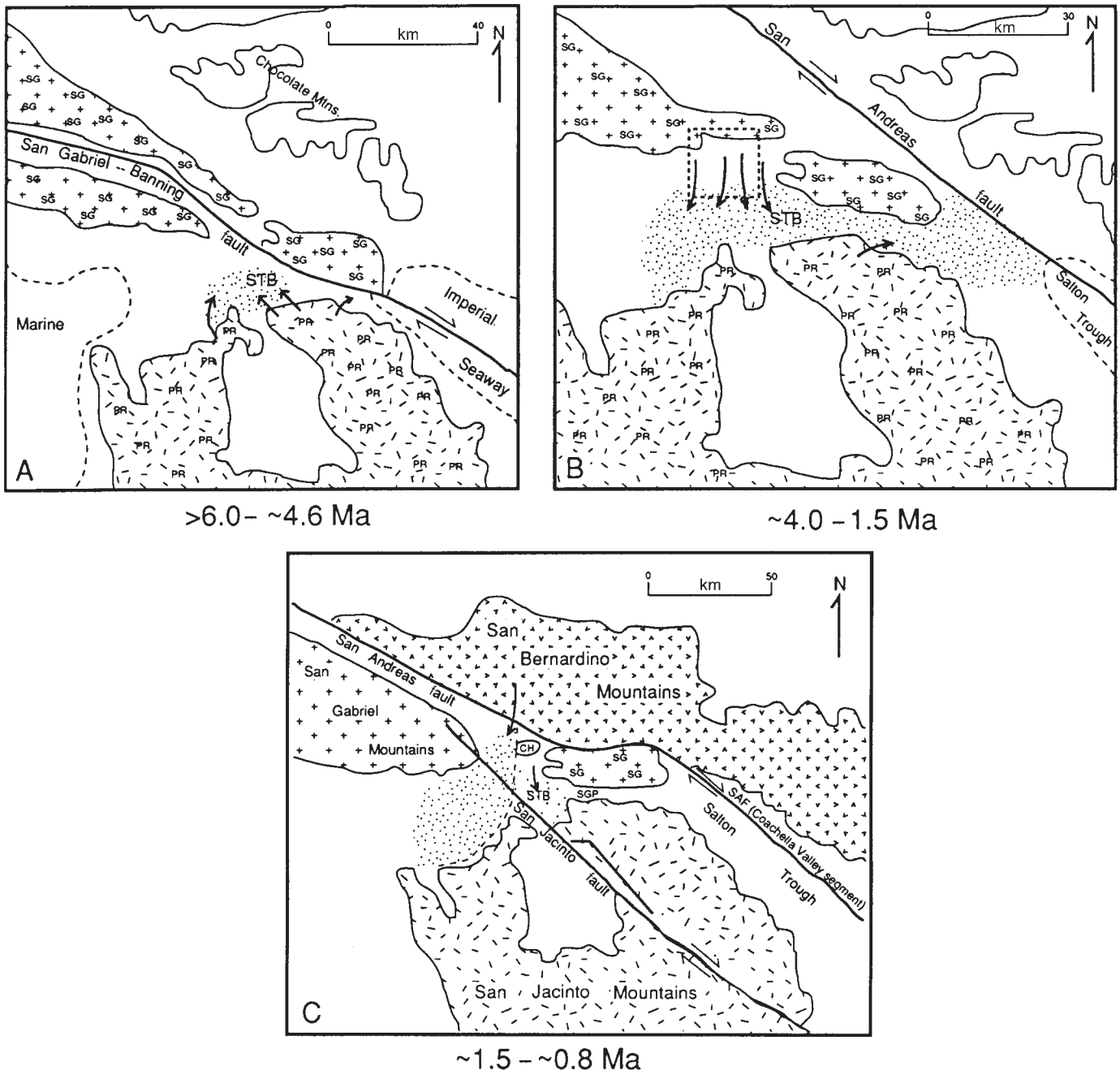


Figure 18. (A) Position of the San Gabriel–Banning fault with respect to the Badlands basin from before 6.0 Ma to about 4.6–4.3 Ma; arrows indicate direction of deposition. The block of San Gabriel–type basement immediately northeast of the Badlands region, and on the opposite side of the fault, was not yet expressed topographically and therefore did not contribute material to the basin. The proto–San Gabriel Mountains to the northwest were still too distant, or not sufficiently uplifted, to contribute. PR—Peninsular Ranges–type basement; SG—San Gabriel Mountains–type basement (modified from Matti and Morton, 1993, Fig. 7E). (B) From ca. 4.0 to 1.5 Ma; San Gabriel–Banning fault has been abandoned in favor of the modern San Andreas fault; the terrane of San Gabriel Mountains–type basement east of the topographically high San Bernardino basin source area (dashed square) is still buried and unable to contribute material to the Badlands sequence (modified from Matti and Morton, 1993, Fig. 7G). (C) Diagrammatic reconstruction at about 1.2–1.5 Ma to about 0.8 Ma. San Bernardino Mountains are sufficiently uplifted to contribute material to the northwesternmost part of the Badlands via an ancestral Santa Ana River drainage; San Gorgonio Pass structural knot has developed with a consequent shift in right-lateral slip from the Coachella Valley strand of the San Andreas fault to a newly formed San Jacinto fault; the terrane of San Gabriel–type basement east of the present-day location of the San Bernardino basin has been uplifted to the extent that it can now provide sediments to the Badlands. CH—Crafton Hills; SGP—San Gorgonio Pass (modified from Matti and Morton, 1993, Fig. 7I); SAF—San Andreas fault; STB—San Timoteo Basin. See Figure 1A for approximate latitude and longitude.

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This major shift in provenance coincides with the generally accepted age of the abandonment of the San Gabriel–Banning fault and activation of the modern San Andreas fault between about 4 and 5 Ma (Powell et al., 1993, and references within) suggesting a causal relationship. However, Weldon et al. (1993, p. 193) found "little evidence for major uplift in the central Transverse Ranges along the San Andreas fault during Pliocene time" and therefore attributed the mostly Pliocene San Timoteo formation to "relief that originated with the late Miocene proto–Transverse Ranges" as a consequence of the then-active San Gabriel fault. As late Miocene uplift progressed, so did right-lateral translation of the Transverse Ranges toward the Badlands depositional area along the San Gabriel–Banning fault. Although uplift and/or lateral translation would result in the depositional pattern seen in the Badlands, i.e., distal deposits at the base represented by the ripple-laminated sandstone, overlain by a progressively coarsening-upward sequence, only translation is temporally consistent with the paradigm of Weldon et al. (1993). No sediments from a Transverse Ranges source appear in the Badlands sequence until the early Pliocene, and the major depositional event that resulted in the bulk of the San Timoteo formation is not apparent until mid-Pliocene time (see following). Thus, the southward progradation of alluvial fan-braided stream deposits from a Transverse Ranges source onto the Peninsular Ranges block between about 4.0 and 6.0 Ma, as hypothesized by Matti and Morton (1993), is here refined to post 4.6 Ma (Fig. 17B).

This may also date abandonment of the San Gabriel–Banning fault. Along Jack Rabbit Trail, unit 3 of the lower San Timoteo formation represents the ~1.5 m.y. following the first appearance of San Gabriel–type clasts (Fig. 17). Although unit 3 is coarser grained than the underlying ripple-laminated sandstone member, its monotonous lithologic character suggests a period of relative tectonic quiescence in the region perhaps related to the abandonment of the San Gabriel–Banning fault. This is consistent with the findings of Weldon et al. (1993), who noted an initial decrease in uplift rates following the shift from the San Gabriel–Banning to San Andreas faults. Abandonment of the San Gabriel–Banning fault would also place the San Gabriel basement source and the San Timoteo basin on the same tectonic block, thus ending any right lateral translation of the former toward the latter (Fig. 18B). However, the major depositional event responsible for accumulation of the bulk of the San Timoteo formation, the thick conglomeratic unit 4, had not yet begun.

The abrupt lithologic change from the generally finer-grained, less conglomeratic unit 3 to the overlying, highly conglomeratic unit 4 signals the beginning of major deposition in the Badlands region. It also calls into question the lack of evidence, noted by Weldon et al. (1993), for uplift in the central Transverse Ranges along the San Andreas fault during Pliocene time. Matti and Morton (1993, p. 146), citing a vertical distribution of clasts in the Badlands sequence "that records progressive unroofing of the San Bernardino and Lytle Creek windows of Pelona Schist during the time frame 4.0 to 2.0 Ma" (Fig. 17A), concluded that during that time, the area of today's San Bernardino Basin, northwest of the Badlands, was a topographic high that provided the source of the San Gabriel–type detritus being shed into the San Timoteo basin (Fig. 18B). That unit 4 persists, and even increases in coarseness stratigraphically upward until another lithologic change occurs at the 1.5 Ma level (Fig. 17B), suggests, in contrast to conclusions of Weldon et al. (1993), that local pre-Quaternary geometric complexities were developing in response to the Pliocene geometry of the San Andreas fault. Unit 4, therefore, documents uplift, active erosion, and the progressive unroofing of the San Bernardino window from at least 4.0 to 1.5 Ma (Fig. 18B).

Deposition of unit 4 also appears to have been time transgressive across the Badlands area from northwest to southeast. Correlation of the base of the Landfill–El Casco section to chron C2Ar implies that deposition of unit 4 had begun by at least 4.0 Ma in that area, whereas in the Jack Rabbit Trail section, about 6 km to the southeast, unit 4 did not become the dominant facies until about 3.1 Ma. This corroborates the proposal of Matti and Morton (1993) that alluvial fan deposits prograded southward across the Badlands area, but also implies that stratigraphic boundaries between units of the San Timoteo formation may be diachronous.

The change near the top of the Landfill–El Casco section from the reddish, conglomeratic sandstones typifying unit 4 below, to grayish, somewhat finer-grained deposits above could be due to tectonic and/or climatic factors. The fossil mammals of the El Casco local fauna support a climatic shift from dryer to wetter conditions. The presence of muskrats and tapirs in the region 1.3–1.4 Ma indicates a permanent water source, and the subtropical to tropical habitus of tapirs today also implies warm, equable conditions. However, Pleistocene tapir fossils are also known from several localities that likely underwent temperate climatic conditions, including Oregon, Nebraska, Pennsylvania, and Ohio (Schultz et al., 1975). The bog lemming, *Mictomys*, on the other hand, is today a sensitive climatic indicator inhabiting only moist areas with cool, cloudy summers (Repenning et al., 1995). Its range has expanded southward or retreated northward with the Canadian forests as glaciation waxed and waned (Repenning and Brouwers, 1992). Absence of the subtropical to warm temperate cotton rat, *Sigmodon*, further suggests cool climatic conditions. Together these faunal criteria, in addition to the change in color and grain size of strata in this part of the section, infer a climate for the Badlands region about 1.3 Ma that included more moisture than today, probably in the form of greater winter precipitation. Summers may have been somewhat wetter than today, but it is unlikely that they were cool, cloudy, and wet, as the Mediterranean climate with its summer drought conditions was emplaced in southern California by post-middle Miocene time (Minnich, 1985; see also Cerling et al., 1998). A generally wetter climate, therefore, may be at least partly responsible for the change in sedimentary facies seen in the upper part of the Landfill–El Casco composite section.

From a tectonic standpoint, the lithologic change coincides with and supports hypotheses regarding the 1.5 Ma inception of the San Gorgonio Pass structural knot and San Jacinto fault, tectonic subsidence of the San Bernardino basin, and uplift of the southern and western fronts of the San Bernardino Mountains (Meisling and Weldon, 1989; Matti and Morton, 1993). Subsidence of the San Bernardino basin would alter deposition into the Badlands and would likely be manifested in the form of a lower energy facies, such as that seen. In contrast, other models suggest an age of about 2.5 Ma for inception of the San Jacinto fault on the basis of a slip rate of ~10 mm/yr over about 25 km of offset (Sharp, 1981; Prentice et al., 1986; Rockwell et al., 1986). It is also at this time that clasts derived from the San Bernardino Mountains first appear in the Badlands sequence (Fig. 17). Conglomeratic deposits from this source occur predominantly in the northwesternmost Badlands region very high in the sequence; they are not found in the part of the sequence overlying the Landfill–El Casco section. Clast suites and paleocurrent data indicate derivation of the San Bernardino material from an ancestral Santa Ana River drainage (Fig. 18C; Morton et al., 1986; D. Morton, 1996, personal commun.). The only age control for these deposits is that provided by a single upper third molar of a primitive species of the meadow vole, *Microtus*, from the Olive Dell local fauna (Repenning, 1987) located "near the top of this conglomeratic section" (Morton and Matti, 1993a, p. 221). Attempts to relocate the original Olive Dell fossil locality were unsuccessful, as were attempts to recover samples for paleomagnetic analysis in the vicinity. However, an estimated age of about 1.5 Ma is based on the following: (1) *Microtus* is not known anywhere in the United States prior to about 1.5–1.6 Ma (Repenning et al., 1995), and (2) the tooth is considerably more primitive than *Microtus* teeth from the <0.99 Ma Shutt Ranch local fauna.

Deposition of San Bernardino Mountains–derived clasts into the Badlands area by an ancestral Santa Ana drainage system would effectively cut off the southward progradation of San Gabriel material from sources north-

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west of that drainage (Fig. 18C). However, clasts in the hundreds of meters of section above the 1.3–1.4 Ma level of the El Casco locality clearly indicate that deposition of San Gabriel-type material had not yet ceased.

The palinspastic reconstructions by Matti and Morton (1993, Fig. 7, A–J) show a terrane of San Gabriel Mountains–type basement immediately north to northeast of the Badlands (Fig. 18, A–C). Today, this region includes those hills east of Crafton Hills, southeast of the present Santa Ana drainage (Figs. 1 and 18C; see Matti and Morton, 1993, Fig. 7, A–J). This terrane must have provided the material for continued deposition of San Gabriel–type clasts coincident with that of San Bernardino clast deposition to the northwest. Why did this terrane not assume a more important role earlier in the depositional history of the Badlands?

The first consideration involves the southeastward progradation of San Gabriel–type material from the San Bernardino basin area from about 4 to 1.5 Ma. If the terrane east of the Crafton Hills was the source of San Gabriel basement–type material into the central Badlands, rather than the San Bernardino basin area, there should be no discrepancy in time of arrival between the Landfill–El Casco and Jack Rabbit Trail sections. The second consideration involves the suite of clasts seen. It is doubtful that the terrane east of Crafton Hills was contributing material to the Badlands from 4 to 1.5 Ma because there are no widespread outcrops of Pelona Schist there that necessarily must be present to account for the distribution of clast types in the Badlands sequence noted by Matti and Morton (1993). This area evidently was not of sufficient topographic expression to undergo active degradation to the extent undergone by rocks to the northwest in the present location of the San Bernardino basin, where clasts of Pelona Schist were available.

J. Matti (1996, personal commun) noted that the overall drainage pattern in this region prior to the structural adjustments caused by juxtaposition of the San Jacinto and San Bernardino blocks was likely toward the Salton Trough based partly on the marine Imperial Formation in San Gorgonio Pass (Fig. 18A). The area in question, therefore, may not have become sufficiently uplifted to allow aggressive degradation until the San Gorgonio Pass structural knot began to develop about 1.5 Ma. Inception of the San Jacinto fault zone, which, according to Morton and Matti (1993a), formed in response to the San Gorgonio Pass knot, resulted in several additional structural adjustments in the region. These adjustments included inception of the San Bernardino basin due to the transtensional geometry of the San Jacinto and San Andreas faults in the area of their merger, and uplift in the San Gorgonio Pass area. It is this uplift that allowed degradation of the San Gabriel basement–type terrane east of Crafton Hills. Although paleocurrent data that might substantiate the drainage pattern are lacking, present-day drainage patterns north of San Timoteo Canyon are oriented northeast-southwest, exactly that required to bring material into the Badlands from this region.

Major deposition in the Badlands did not cease until a time postdating the youngest dated fossil assemblage in the San Timoteo formation, the Shutt Ranch local fauna. Morton and Matti (1993a, p. 224) stated, "[t]he time involved in the deposition of the part of the San Timoteo Formation overlying the beds of the Shutt Ranch local fauna is estimated at 0.1 to 0.3 Ma." This resulted in an estimate for the end of deposition of "about 1 ± 0.3 Ma" based on an estimated age of the fauna of 1.3 ± 0.3 Ma. In this report, it was determined that the age of the Shutt Ranch assemblage is between 0.99 and 0.78 Ma. If estimates for the time of deposition of strata overlying the Shutt Ranch locality are accurate, then deposition in this part of the Badlands did not cease until ca. 0.8 to 0.5 Ma. Thus, the shift in the Badlands from an aggradational system to one of erosion and degradation is constrained to a time postdating about 0.8 Ma. It is noteworthy that these estimates are consistent with similar estimates for the age of the oldest geomorphic surface in the Badlands based on stage of soil development (K. Kendrick and J. Matti, 1996, personal commun.; Kendrick et al., 1993; Kendrick and McFadden, 1996).

CONCLUSIONS

The magnetic reversal pattern in the San Timoteo Badlands sequence, in concert with biochronological information, including Hemphillian, Blancan, and Irvingtonian mammal assemblages, indicate correlation to chron C3An.2n near the base of the section, at about 6.3 Ma, to within the Brunhes normal chron near the top of the section, after 0.78 Ma. This chronologic framework is the first in North America developed from a single stratigraphic sequence that extends across these three land mammal ages. It provides refined age estimates for the previously known, but poorly constrained Mount Eden, El Casco, and Shutt Ranch local faunas, and for newly discovered Blancan assemblages. The Mount Eden local fauna is dated as ca. 5.6 Ma; the El Casco local fauna is ca. 1.4 Ma; and the Shutt Ranch local fauna is no older than 0.99 and perhaps younger than 0.78 Ma.

The chronologic framework developed here is also consistent with, but refines and strengthens, previous models that address the timing of large-scale tectonic events in the Transverse Ranges region. From before 6.3 Ma until about 4.6 Ma, the Badlands area was under the influence of deposition from a Peninsular Ranges basement source to the south that ultimately resulted in accumulation of the fluvio-lacustrine deposits that typify the Mount Eden formation. After about 4.6 Ma, deposition of the San Timoteo formation began as the provenance shifted to a source north of the Badlands in the eastern San Gabriel Mountains and the present-day locality of the San Bernardino basin. The generally coarsening-upward nature of the San Timoteo formation suggests the progressive uplift of the source area, or the gradual translation of the source area toward the Badlands basin along a right lateral San Gabriel–Banning fault during latest Miocene–earliest Pliocene time. However, the major phase of deposition of the San Timoteo formation began during mid-Pliocene time, and lasted until about 1.5 Ma, suggesting renewed uplift in the region as a consequence of the shift from the San Gabriel–Banning fault to the modern trace of the San Andreas fault zone in southern California. Although this study does not provide a refined age estimate for the earliest deposition of San Bernardino Mountains–type clasts in the Badlands sequence, reevaluation of the few data still supports a date no older than about 1.5 Ma. As San Bernardino Mountains material was deposited in the northwesternmost parts of the Badlands area by an ancestral Santa Ana River, deposition of San Gabriel–type material in the southeastern part, from a source area to the east of Crafton Hills, continued until a time postdating the 0.78 to 0.99 Ma Shutt Ranch local fauna. After that time, the Badlands region was dominated by erosion and degradation rather than aggradation.

Previous studies have found that tectonic vertical-axis rotation within regions of right-lateral shear, including the San Jacinto fault zone, occurs in a clockwise fashion. Results of this study, however, indicate that counterclockwise rotation may have occurred in the Badlands region after about 4.5 Ma, although earlier suggestions of ~20° may be exaggerated and more a result of magnetic overprinting and poorly understood structural complications than tectonics. It is hoped that this work will stimulate further investigations to help resolve these problems.

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